



Review

Marine antifouling strategies: Emerging opportunities for seawater resource utilization

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ABSTRACT

The attachment and colonization of fouling organisms, including bacteria, algae, and invertebrates onto submerged surfaces is a widespread issue known as marine biofouling. This problem has substantial implications for various aspects such as economics, environment, and ecology. The prohibition of traditional antifouling methods, such as those based on tributyltin, has been enforced due to their accumulation and high levels of toxicity. Consequently, there is an urgent demand for the development of antifouling coatings that are both environmentally friendly and effective. The formation and detrimental effects of biofouling, the historical progression of antifouling technologies, and the mechanism of modern biofouling control are introduced. This review encompasses recent advancements in marine antifouling strategies, including bioactive antifoulant mechanisms, self-polishing antifouling mechanisms, surface wettability mechanisms, photocatalytic bactericidal mechanisms, and biomimetic antifouling mechanisms. Based on the strategies described above, the antifouling systems employed in the extraction of uranium and lithium from seawater, as well as net cages for aquaculture are explored. It is expected that marine antifouling technologies will provide more opportunities for the system design of seawater resource utilization.

1. Introduction

Marine fouling organisms refer to a broad range of organisms, including animals, plants, and microorganisms, that attach to and inhabit ships and marine engineering facilities in the marine environment, resulting in economic losses and environmental safety risks. The metal corrosion and the reduced service life of ships and offshore buildings are accelerated by the damage caused to the anti-corrosion layer by marine fouling organisms. Biofouling also increases hydrodynamic resistance, reduces vessel maneuverability, and increases fuel consumption [1]. Additionally, fouling organisms also present serious challenges for marine ecosystems, increased greenhouse gas emissions, and the transregional migration of alien species using the hull as a carrier [2]. Given that the oceans encompass approximately 70 % of the earth's surface area and that marine transport accounts for 90 % of global trade, marine resources, and marine industries have become indispensable pillars of economic development [3]. Therefore, the development of antifouling materials and protective technologies has

become a worldwide issue [4].

The phenomenon of biofouling, which entails the accumulation of various organisms on artificial surfaces upon exposure to seawater, is a pervasive issue [5]. The process of formation is distinguished by four primary stages. In the first stage, there is rapid aggregation of organic molecules (such as proteins and polysaccharides) on the surface, leading to the creation of a conditioning film. The process is primarily controlled by physical effects such as Brownian motion, electrostatic forces, and van der Waals forces [6]. The second stage is the rapid attachment of bacteria and unicellular diatoms to the surface of the conditioning film, ultimately resulting in the formation of primary film. In the third stage, the attached microorganisms secrete various substances, including polysaccharides, proteins, lipids, and nucleic acids to induce more fouling organisms such as crustacean larvae, marine bacteria, and protozoa, thereby creating a highly complex community. The fourth stage involves the attachment and proliferation of marine invertebrates and macroalgae, which belong to macrofouling [7]. Despite the apparent linearity of the fouling process, wherein each stage is dependent on the

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preceding one, it is not strictly sequential. In other words, the absence of fouling organisms in one stage does not necessarily impede the progression of subsequent stages. See (Table 1).

The early antifouling method was to use heavy metals to kill fouling organisms. Copper was widely used in hull protection as an antifouling agent. However, given the existence of more than 4,000 different fouling species, copper is hardly effective against all fouling organisms [8,9]. In the mid-20th century, the powerful antifouling (AF) effect of tributyltin (TBT) was discovered, which led to TBT becoming the primary biocide component of AF paint products [10]. However, TBT exhibits high toxicity towards a wide range of non-target aquatic organisms, even at low concentrations. Furthermore, the propensity of TBT compounds to accumulate in organisms raises concerns regarding potential human health risks associated with the consumption of contaminated seafood. In response, a series of national regulations have been implemented globally to restrict the use of TBT compounds. However, because of the cross-border nature of maritime transport, these regulations have not yielded the intended outcomes. Until 2008, the International Maritime Organization banned the use of TBT as an AF compound [11]. To compensate for the shortage of traditional antifouling coatings, there has been an increase in the research of novel antifouling coatings both domestically and internationally in recent years. Modifying the traditional antifouling coating, selecting environmentally friendly raw materials, or modifying the raw materials, not only can improve the antifouling ability of the coating itself but also can promote some functional combinations to improve the performance of the antifouling coating, to adapt to the complex marine environment and reduce

environmental pollution [12–14].

Seawater is a valuable resource with abundant reserves of natural uranium (around 4 billion tons) and lithium (approximately 230 billion tons) [15,16]. Additionally, it provides a favorable environment for aquaculture. However, the utilization of seawater resources faces several challenges. One such challenge is the negative impact of fouling organisms on the adsorbents used in seawater uranium extraction. This can significantly degrade the performance of uranium extraction and reduce the reusability of the adsorbents. Similarly, membrane separation technology and adsorption methods employed for lithium extraction from seawater are also vulnerable to fouling organisms, leading to a decrease in the efficiency of extraction. Aquaculture nets, due to their large surface area and the presence of residual bait, feces, secretions, and other nutrients from the aquaculture process, promote the attachment and growth of fouling organisms on aquaculture facilities. This additional weight causes deformation and distortion of the fishing nets. Hence, the development and application of anti-fouling materials have emerged as a promising approach to mitigate the adverse effects of marine fouling organisms on the utilization of seawater resources.

At present, substantial literature has been published regarding the marine antifouling system, and the classification of antifouling coating appears to be increasingly refined. However, previous literature has focused on antifouling coatings *per se* and has not provided a comprehensive review of recent advancements in antifouling systems for the utilization of seawater resources [17–19]. This paper provides an overview of combination of antifouling strategies for marine equipment and seawater resource utilization. It covers various strategies including bioactive antifoulant mechanisms, self-polishing antifouling mechanisms, surface wettability mechanisms, photocatalytic bactericidal mechanisms, and biomimetic antifouling mechanisms. Additionally, the paper highlights the combination of antifouling systems with marine resource applications, particularly in the extraction of uranium and lithium from seawater as well as net cages for aquaculture. Fig. 1 gives an overview of antifouling strategies for marine equipment and seawater resource utilization.

Table 1
The advantages and disadvantages of different antifouling strategies.

Antifouling strategies	Advantages	Disadvantages	Ref.
Bioactive antifoulant mechanisms	Easy to prepare and low cost. Good antifouling performance.	Potential impacts on non-target organisms and marine ecosystems are uncertain. Solubility, stability and release pattern of the antifouling agent in seawater need to be considered.	[94]
Self-polishing antifouling mechanisms	Extending antifouling period through self-polishing behavior and controlled biocide release rates.	High production costs. Limited durability, Limited antifouling performance in static conditions. Difficult to hydrolyse the copolymer backbone of most materials.	[95,96]
Surface wettability mechanisms	Environmentally friendly. Good expansibility.	Poor adhesion strength and weak durability. Limited antifouling performance in static conditions.	[97,98]
Photocatalytic bactericidal mechanisms	Reusable. Environmentally friendly.	Limited control of seawater temperature, pH, dissolved oxygen and light intensity. Limited large-scale practical application.	[80,84]
Biomimetic antifouling mechanisms	Environmentally friendly. Good antifouling performance under laboratory conditions.	Further research is needed to fully understand the principles of certain bionic antifouling coatings. Limited large-scale practical application.	[91,93]

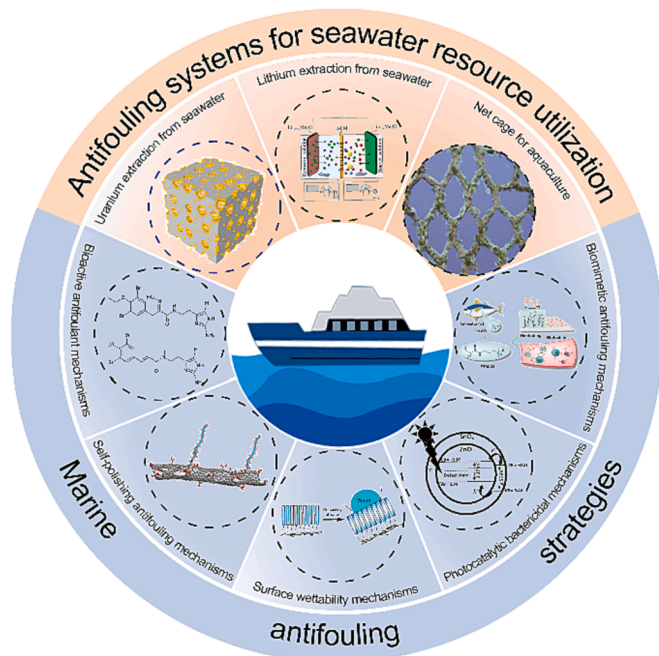


Fig. 1. Overview of marine antifouling strategies and antifouling systems for seawater resource utilization.

2. Marine antifouling strategies

2.1. Bioactive antifoulant mechanisms

Bioactive antifouling agents exert their functions by making contact with fouling organisms. The inhibition effect, the microbicidal effect, and the action mechanism of these antifoulants are variable across different antifouling agents. Antifouling agents can be categorized into two distinct groups: natural agents, which are isolated from living organisms, and synthetic agents, which are artificially produced through chemical synthesis. The chemical structure of the 15 antifouling agents discussed in this section (hereinafter designated as Compound 1–15) is shown in Fig. 2.

2.1.1. Natural antifouling agents

In light of the limitations of traditional antifouling agents, a viable alternative is to find environmentally benign organic antifoulants as substitutes. The ideal antifoulants should possess broad-spectrum and long-term effectiveness, while also exhibiting low toxicity to non-targeted organisms, non-bioaccumulation and degradation. Numerous natural compounds from living organisms showed these favorable properties and possessed better biocompatibility and degradability. Therefore, natural antifoulants have become a promising research field [20]. For example, antibiotics, as one of the important products of secondary metabolism, can inhibit the growth of other microorganisms even at low concentrations [21].

In the natural environment, many organisms possess the ability to secrete antibacterial biopeptides, such as alkaloids, organic acids, terpenoids, and amides, which effectively inhibit the adhesion of fouling organisms. Some marine organisms have been found to produce specific metabolites that suppress the activity of fouling organisms. Notably, marine invertebrates and algae exhibit resistance to fouling organisms by producing and releasing low-toxic bioactive substances within their physiological systems. Researchers identified several natural antifouling agents from various organisms, such as coral and algae, which have the potential to be used in the development of novel biomimetic antifouling

coatings [22,23]. Wang et al., isolated several cypress diterpene active substances from the soft coral *Sinularia flexibilis* by Toyopearl HW-40 column chromatography and High-Performance Liquid Chromatography (HPLC). Among them, (–)-14-deoxycrassin (Compound 1) exhibited excellent performance against the two fouling organisms. The EC₅₀ (median effective concentration) value for *B. neritina* was 3.90 µg/ml, while the EC₅₀ value for *B. albicostatus* was 21.26 µg/ml. Additionally, it exhibited low toxicity towards *B. albicostatus* larvae with an LC₅₀ (median lethal concentration) value greater than 100 µg/ml [24]. Levert et al., conducted a study to evaluate the activities of five meroterpenoids derived from the Ascidian *Aplidium aff. densum*. The aim of their investigation was to evaluate the potential inhibitory effects on growth exhibited by six different strains of marine bacteria that were isolated from various surfaces in the marine environment. The MIC values of cordiachromene A (Compound 2) were determined to be 125 µg/mL for Gram-positive bacteria (*Bacillus* sp., *B. firmus* and *Actinobacteria*), while for Gram-negative bacteria (*Roseobacter* sp., *Pseudomonas* sp., *P. nigrificiens*), it was determined to be 32 µg/mL. In the case of Epiconicol (Compound 3), its MIC values against *Bacillus* species were found to be 63 µg/mL, whereas against other species, it showed a MIC value of 32 µg/mL [25].

In addition to releasing bioactive substances, sponges and their associated microbiota have been found to synthesize compounds that disrupt quorum sensing (QS) mechanisms [26,27]. Quorum sensing (QS) serves as a synchronization mechanism employed by bacterial populations, enabling communication, behavior regulation, and assessment of population density. The utilization of marine biological secretions to disrupt bacterial quorum sensing effectively impedes the formation of bacterial biofilms, thereby impeding the attachment of organisms [28]. Tintillier et al., discovered four novel brominated tyrosine metabolites (Compound 4–7) from the French Polynesian sponge *Pseudoceratina n. sp.* Additionally, they discovered two 2-aminoimidazolic derivatives (Compound 8 and 9) from the sponges *Psammaphysilla purpurea* and *Verongula* sp. Compound 8 demonstrated exceptional growth inhibition against *V. aestuarianus*, displaying an MIC of 0.01 µM. Compounds 4–9 demonstrated promising quorum sensing inhibition (QSI) and possessed

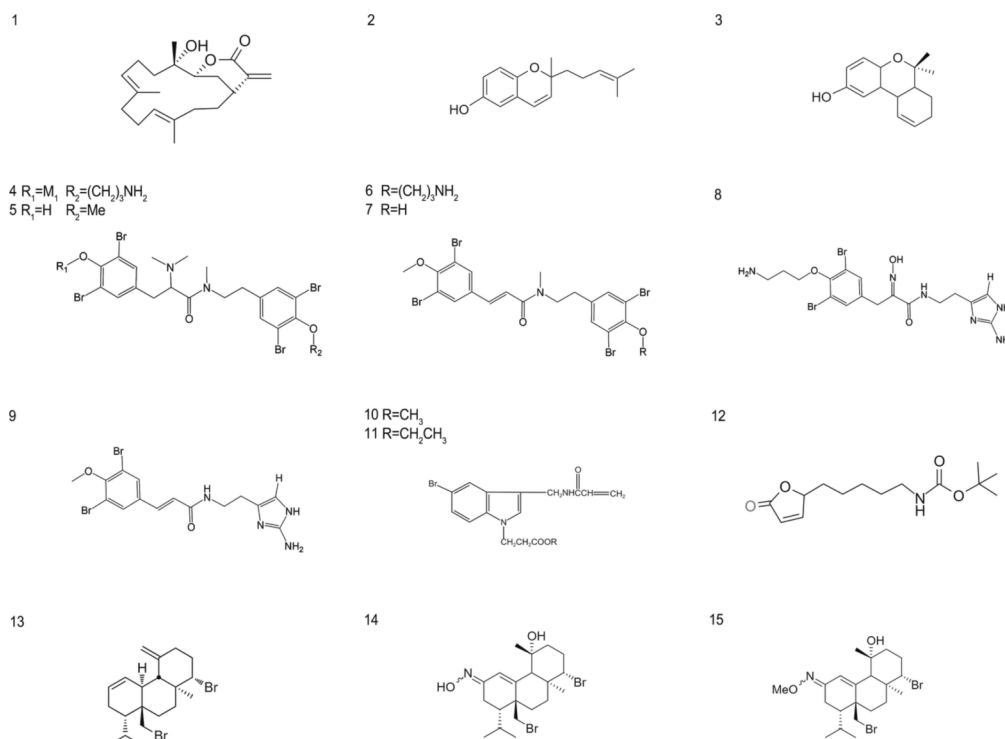


Fig. 2. The molecule structures of antifouling agents.

antifouling activities against various strains of bacteria and microalgae [29].

2.1.2. Synthetic antifouling agents

However, the industrial production of natural antifouling agents remains challenging due to the complex extraction and low yield. Furthermore, the extensive implementation of these applications may impose a risk to the existence of the origin organisms and detrimentally impact the balance within the ecosystem [30]. Therefore, researchers have attempted to enhance the efficacy and non-toxic antifoulants through a simplified and controlled synthetic process by gaining a comprehensive understanding of antifouling mechanisms and potential conformational relationships of natural antifoulants. For large-scale applications, this approach demonstrates viability. Numerous compounds with favorable antifouling effects have been synthesized and examined thus far. The findings indicate that certain synthetic analogs are more effective in fouling organisms and less toxic to non-targeted organisms than natural antifouling agents. Consequently, synthetic antifouling agents hold greater promise for practical applications, encompassing isothiazolinones, indoles, furans, resins, pyridines, and dithionite [31]. Ni et al., synthesized two novel indole derivatives containing acrylamide groups (Compound 10 and 11). These compounds were synthesized using the Michael addition reaction and Friedel-Crafts reaction with 5-bromo-1H-indole as the raw material. When tested against a panel of bacteria, both Compound 10 and Compound 11 exhibited remarkable efficacy, surpassing that of chlorothalonil. Evaluation of their MIC (mg/mL) values demonstrated potent antibacterial activity. Moreover, the anti-algal assay exhibited substantial inhibition rates (over 70 %) against *C. vulgaris*, *N. clostridium*, and *S. costatum*, further emphasizing the potential of these compounds [32]. Chiang et al., modified a butenolide derivative into Boc-butenolide (Compound 12). *A. amphitrite* and *H. elegant* larval settlement bioassay results demonstrated that Boc-butenolide exhibited effective antifouling ability against macro foulers (the rates of settlement reaching approximately 70 % at a concentration of 3.125 ppm) [33]. Prousis et al., conducted a synthesis of bromophenol derivatives that were derived from brominated diterpenes isolated from the red alga *Sphaerococcus coronopifolius*. The settlement inhibition assay against *A. Amphitrite* showed Compound 14 and 15 have promising antifouling efficacy ($EC_{50} < 0.5$ mg/L), and Compound 13 and 14 showed $LC_{50}(\text{cypris})$ values higher than the maximum tested concentration (50 mg/L) in cyprids mortality [34].

In summary, it can be argued that both natural and synthetic antifouling agents possess environmental advantages over conventional metal AF agents. These alternatives are characterized by their eco-friendliness, biodegradability, cost-effectiveness, and potential utilization as precursors in the development of antifouling coatings. However, it is important to note that only a small number of them have demonstrated potential in practical production applications. In addition, it should be acknowledged that a single antifouling agent exhibits antifouling properties exclusively against specific fouling organisms, rendering it incapable of achieving broad-spectrum antifouling efficacy. In complex marine environments, the potential adverse impacts of antifouling agents on non-target organisms and oceanic ecosystems remain uncertain. The solubility, stability, and release rate of antifoulants in seawater also need to be considered, which becomes critical for the lifetime of antifoulants. Therefore, natural antifoulants still need further research in marine antifouling, and future antifoulants must synergize with other antifouling strategies to effectively prevent and control fouling organisms.

2.2. Self-polishing antifouling mechanisms

The acrylic or methacrylic copolymers are used as the polymer matrix of self-polishing antifouling technology. In the presence of weakly alkaline seawater, these copolymers undergo gradual hydrolysis,

resulting in an increase in hydrophilic groups along the main chain. Consequently, the outermost layer of the copolymers continuously peels off, preventing fouling organisms from adhering to the surface. This mechanism effectively achieves a desirable antifouling effect [35]. Initially, self-polishing coatings predominantly contained TBT, which exterminated marine fouling organisms under its toxicity. Therefore, the traditional self-polishing coatings with TBT exhibited excellent antifouling properties under the combined effect of both. However, researchers have demonstrated that TBT can induce variations in marine organisms and reduce their diversity. As a result, the International Maritime Organization has explicitly prohibited its utilization, leading to a rise in the prominence of tin-free self-polishing antifouling coatings. This section presents synergistic mechanisms of self-polishing and bioactive agents, and degradable self-polishing mechanisms.

2.2.1. Synergistic mechanisms of self-polishing and bioactive agents

The traditional self-polishing coatings with TBT possess promising antifouling performance due to the synergistic effect of self-polishing action and biocidal properties of TBT. Therefore, the introduction of bioactive agents is an approach to improve the antifouling property. The bioactive agents are generally introduced to the side chain of the copolymer. This copolymer can be released into the seawater through hydrolysis, resulting in a constant renewal surface effect. Additionally, the bioactive agents incorporated into copolymers can play a role in biofouling control.

Bioactive agents can be classified into two categories: organic and inorganic. The application of antifouling agents on ship hulls is a historically significant technology. However, partial inorganic bioactive agents such as lead, arsenic, mercury, and their organic derivatives have been prohibited due to their environmental hazards [36]. Copper material is currently regarded as the most significant inorganic bioactive agent due to its low cost, excellent antifouling performance, and low toxicity (approximately $1/40 \sim 1/50$ of organic tin). Consequently, it serves as an ideal alternative to organic tin [37]. The widely used antifouling coating consists of a Cu_2O antifouling agent. The agent generates copper ions in seawater, which leads to the deactivation of the primary enzymes that marine organisms rely on for survival. Consequently, this process induces changes in biological tissues and ultimately causes the death of these organisms [38]. Ding et al., prepared self-polishing coatings of Cu_2O /copolymer/polycaprolactone (PCL) by atom-transfer radical polymerization (ATRP) method. The copolymer was composed of poly(lauryl methacrylate)-b-poly(2-(N,N-dimethylamino)ethyl methacrylate) (PLMA-b-PDMAEMA). The thickness of the hydrolyzed films containing copolymers with 240 and 320 repetitive units (PLMA and PDMAEMA) is comparable to that of the pure resin. The films with $\text{PLMA}_{160}\text{-b-PDMAEMA}_{80}$ and $\text{PLMA}_{213}\text{-b-PDMAEMA}_{107}$ coatings showed a remarkable ability to eliminate over 99.0 % of *S. aureus*, which can be attributed to the enhanced hydrophobic interaction between the copolymer and PCL resin, as well as the improved Cu^{2+} -retaining capacity of the films (Fig. 3a and b) [39].

Quaternary ammonium compounds have been extensively investigated as synthetic bioactive agents and have demonstrated favorable AF properties. These compounds adhere to negatively charged cell surfaces through electrostatic attraction, leading to modifications in the permeability and integrity of the cell membrane, ultimately resulting in the inactivation of enzymes in bacteria. Furthermore, the inclusion of longer alkyl chains in quaternary ammonium compounds enhances their binding affinity to cells. Increasing the molecular weight of quaternary ammonium compounds can effectively increase the density of positive charges in their salt molecules, thus enhancing their AF properties against microorganisms [40–43]. Wang et al. reported dimethyl octadecyl [3-(trimethoxysilyl) propyl] ammonium chloride was tethered to SiO_2 nanoparticles to produce quaternary ammonium salt- SiO_2 (QAS- SiO_2) and further blended them with self-polishing polymer (SP). The self-polishing rate went up as the triisopropylsilyl methacrylate (TISM) content in the polymer increased, in line with expectations The

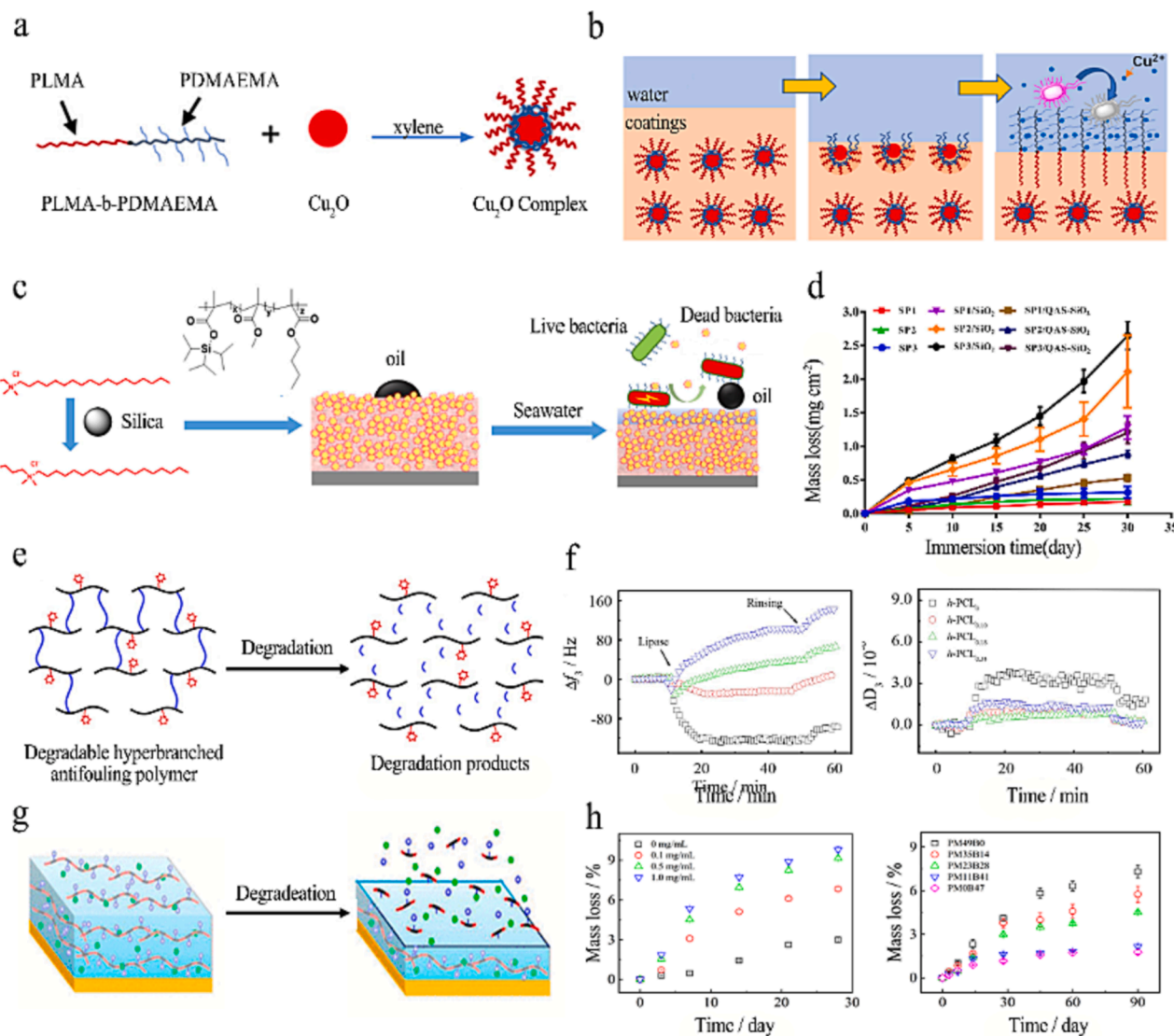


Fig. 3. (a) Formation of Cu_2O microcapsules. (b) Antifouling mechanism of Cu_2O /PLMA-b-PDMAEMA/PCL coatings in seawater [39]. Copyright 2022. Adapted with permission from Elsevier. (c) The antifouling mechanism of QAS-SiO₂ incorporated self-polishing coatings in seawater. (d) The mass loss of SP_x, SP_x/SiO₂ and SP_x/QAS-SiO₂ coatings in artificial seawater [44]. Copyright 2020. Adapted with permission from Elsevier. (e) Schematic of degradation of DHAP. (f) The time-dependent changes in frequency shift (Δf) and dissipation shift (ΔD) during enzymatic degradation of DHAP with varying CL content in artificial seawater at 25 °C [45]. Copyright 2021. Adapted with permission from Elsevier. (g) Depicting the degradation mechanism of PM_xBy. (h) Investigating the mass loss of PM₂₃B₂₈ in different lipase concentrations in artificial seawater at 25 °C, and exploring the mass loss variations of PM_xBy in artificial seawater at 25 °C [46]. Copyright 2019. Adapted with permission from the American Chemical Society Publications.

sterilization rate of SP/QAS-SiO₂ coating is up to 91.5–97.4 %, which is much higher than SP coating and SP/SiO₂ coating (Fig. 3c and d) [44].

2.2.2. Degradable self-polishing mechanisms

With regard to degradable self-polishing antifouling coatings, the design enables surface renewal through main-chain degradation and side-chain hydrolysis. This design feature facilitates a spontaneous self-polishing process that does not rely on water flow. By diminishing the duration of contact between organisms and the surface, the interaction between them is minimized, resulting in reduced adhesion and interaction forces. As a result, fouling organisms encounter difficulties in adhering to the surface. Additionally, these copolymers can not accumulate in the marine environment due to the main-chain degradation and side-chain hydrolysis.

Ai et al., [45] synthesized a degradable hyperbranched antifouling polymer (DHAP) by reversible addition-fragmentation chain-transfer (RAFT) polymerization. The degradation rate of DHAP is significantly affected by the content of caprolactone (CL). Furthermore, the

dissipation (ΔD) generally exhibited an upward trend with increasing thickness but showed a decline when the layer's rigidity is considered (Fig. 3e and f). The antimicrobial performance tests demonstrated that DHAP samples containing 2 wt% Econeal exhibited a lower number of bacterial colonies. Dai et al., [46] synthesized a degradable poly(ester-co-acrylate) copolymer with antifoulant pendant groups using radical ring-opening polymerization. The copolymer is designated as PM_xBy, where x and y represent the weight percentages of N-methacryloyloxy methyl benzo isothiazolinone (V-BIT) and polyester, respectively. After 90 days, the mass loss percentages for PM₁₁B₄₁, PM₂₃B₂₈, PM₃₅B₁₄, and PM₄₉B₀ were found to be 2.1 %, 4.5 %, 5.8 %, and 7.3 %, respectively. This can be attributed to an increase in the main chain ester bond content and a decrease in the BIT side group content. Therefore, the hydrolysis rate of the coating can be controlled by adjusting the main chain ester content (Fig. 3g and h).

In conclusion, self-polishing antifouling mechanisms represent considerable potential for biofouling control due to the self-renewal ability of the copolymer surface. However, it also has the following

drawbacks and areas for enhancement: (1) The high production cost of self-polishing antifouling mechanisms poses a constraint on their widespread implementation. (2) While the antifouling process effectively reduces the attachment of fouling organisms by self-polishing and degradation action, its durability is limited, necessitating frequent reapplication. (3) The mechanism is predominantly used on the rigid substrates like steel plates and ceramics, posing challenges when attempting to apply them on intricate materials such as soft surfaces [47]. (4) The stable polishing rate is a major challenge during development. (5) The conventional self-polishing mechanism plays a role in the biofouling control, relying on the shear force from seawater. (6) The post-hydrolysis copolymers may result in the accumulation of microplastics due to the non-degradable main chain.

2.3. Surface wettability mechanisms

The physicochemical process of biofouling formation on surfaces is intricate and multifaceted. The surface properties of materials, encompassing surface charge, free energy, morphology, and wettability, exert significant influences on the adherence of fouling organisms. Previous studies have demonstrated that the wettability of solid surfaces is determined by two factors: surface chemical composition and surface microstructure [48]. The measurement of solid surface wettability relies on four crucial parameters: contact angle, surface free energy, sliding angle [49] and underwater oil adhesion force [50]. The measurement of solid surface wettability relies on three crucial parameters: contact angle, surface tension, and surface free energy. Surface wettability mechanisms have always exhibited favorable anti-biofouling and self-cleaning properties [51]. This section focuses on the three primary types: hydrophobic, hydrophilic, and amphiphilic antifouling mechanisms.

2.3.1. Hydrophobic antifouling mechanisms

Nowadays, antifoulants are used as one of the functional materials of antifouling strategies. Dependence solely on antifouling agent mechanisms for combating biofouling is insufficient in terms of antifouling efficacy. It is important to explore additional antifouling strategies that synergistically enhance fouling resistance. For example, one approach involves establishing a weak connection between the surface (predominantly hydrophobic surface) and the biological matter. Consequently, the fluid shear force generated by the movement of the ship aids in removing the settled organisms from the surface, thus reducing the attachment of fouling organisms. Therefore, it is crucial to maintain a high level of hydrophobicity on the surface. Achieving this necessitates a combination of reducing the surface free energy and creating microstructures [52]. Despite the general notion that lower surface free energy usually corresponds to a weaker capacity for spontaneous interaction with other compounds and materials, the Bayer curve reveals that the minimum level of bioadhesion does not coincide with the lowest critical surface energy [53]. It is generally believed that it has an excellent antifouling and desorption cleaning effect when the surface free energy is between 20–30 mJ/m².

To guarantee the enduring stability of marine hydrophobic coatings, it is imperative to take into account not only the surface's physical and chemical features, but also aspects like the thickness of the coating, their mechanical stability, and their adhesion to the ship surface. Numerous contemporary techniques for fabricating hydrophobic materials involve the utilization of organosilicon compounds, which are closely linked to the structure and properties of organosilicon [54]. The majority of organosilicons encompass hydrophobic groups similar to $-C_nH_{2n+1}$, thereby enabling their monomers to yield surfaces with low surface energy, or alternatively, they can be modified to enhance hydrophobicity [55]. Organosilicon materials also possess the capability to generate nanostructures via dehydration condensation, or serve as carriers to provide adsorption sites for other materials, thereby facilitating the attainment of sufficient surface roughness for hydrophobic surfaces.

Additionally, the majority of organosilicon reactions occur under mild conditions, allowing for their completion under both acidic and alkaline environments, thereby rendering them convenient for the preparation of hydrophobic materials [56]. Liu et al., prepared silver nanoparticles (AgNP) hybrid organosilicon antifouling coatings (AgNPs-X%/PDMS-DOU) with self-healing and long-term antifouling properties. The compositions of the coatings included double-ended hydroxypropyl polydimethylsiloxane (PDMS), polytetrahydrofuran (PTMG), 2-hydroxyethyl disulfide (HEDS), dimethylglyoxime (DMG), CuCl₂, and AgNPs. As the content of hydrophobic AgNPs increased, the water contact angle of AgNPs-X%/PDMS-DOU gradually increased from $92.63 \pm 2.2^\circ$ to $104.82 \pm 2.1^\circ$. Antifouling tests showed that the control sample, used as a reference, had a high-density bacterial adhesion. As the content of AgNP increased, the antibacterial efficiency was observed to reach 97.8 % [57].

The organic fluoro-based antifouling material possesses several notable attributes, including large electronegativity, small radius, short C-F bond, and high bond energy. The repulsion between fluorine atoms within the polymer engenders stability of the fluorine atoms surrounding the carbon chain, thereby yielding a surface with low surface free energy, strong hydrophobicity, and certain oleophobicity. Therefore, this characteristic can serve as one of the fouling-release mechanisms. The silicon-based and fluoro-based mechanisms differ as follows. (1) The surface free energy of organic fluoropolymer is lower than that of organosilicon polymer. (2) The rigidity of fluoro-carbon chain restricts the molecular chain flexibility of organic fluoropolymer, resulting in a higher modulus compared to organosilicon polymer. Consequently, a greater amount of energy is required to remove marine fouling organisms from the surface of organic fluoropolymer. (3) The cost of fluorinated polymer exceeds that of silicone polymer, and the preparation process is complex. (4) The degradation of long halothane compounds poses challenges in natural processes, while their accumulation in the food chain is facilitated [58]. Therefore, it is important to develop an environmentally friendly fluoropolymer that can be easily processed. Recently, the development of fluorine-containing antifouling mechanism has concentrated on structures with short fluoride side chains, such as fluorinated polyoxetane resins. This choice is attributed to the polymer's characteristics, including a low glass transition temperature, strong adhesion, excellent mechanical properties, and low surface free energy. Cui et al., designed a novel UV/moisture dual-curing hybrid coating consisting of two fluorinated polyoxetane resins. As the ratio of isocyanatopropyltrimethoxysilane (IPTS) end-capped P3F increased, the contact angle of the hybrid coating exhibited a steady increment from 100° to 110° . Simultaneously, the surface energy of the coating gradually decreased from 22.6 mN/m to 18.2 mN/m [59]. Wu et al., synthesized an optimized fluorinated waterborne polyurethane (FWPU) resin with enhanced surface hydrophobicity and antifouling properties via a thiol-ene 'click' reaction (Fig. 4a). Upon reaching a fluorine content of 2 %, the water contact angle (WCA) of FWPU films exhibited a sudden increase from $67 \pm 2^\circ$ to $91 \pm 3^\circ$ (Fig. 4b). The migration of fluorine units to the air/polymer interface of the FWPU films is predominantly responsible for this notable transformation, which subsequently prompts the creation of a denser fluorinated layer. As a result, the surface free energy is diminished, and the water contact angle (WCA) experiences an increase [60].

In addition, numerous wettability-related phenomena observed in nature, such as the self-cleaning effect of lotus leaves and cicada wings, as well as the hydrophobicity conferred by the otter leg surface, can be attributed to the distinctive micro-nano structure of these surfaces. The special wettability of the functional surface is achieved through the synergistic action of chemical composition and the micro and nano-structure present on the surface [61]. Inspired by the antifouling strategy of soft coral (garland fleshy soft coral), Wei et al., chose silicone rubber (RTV-2), whose hydrophobicity is similar to coral skin, as the base material. The hydrophobic composite structure of the upper transparent layer and the lower porous layer was prepared. Both layers

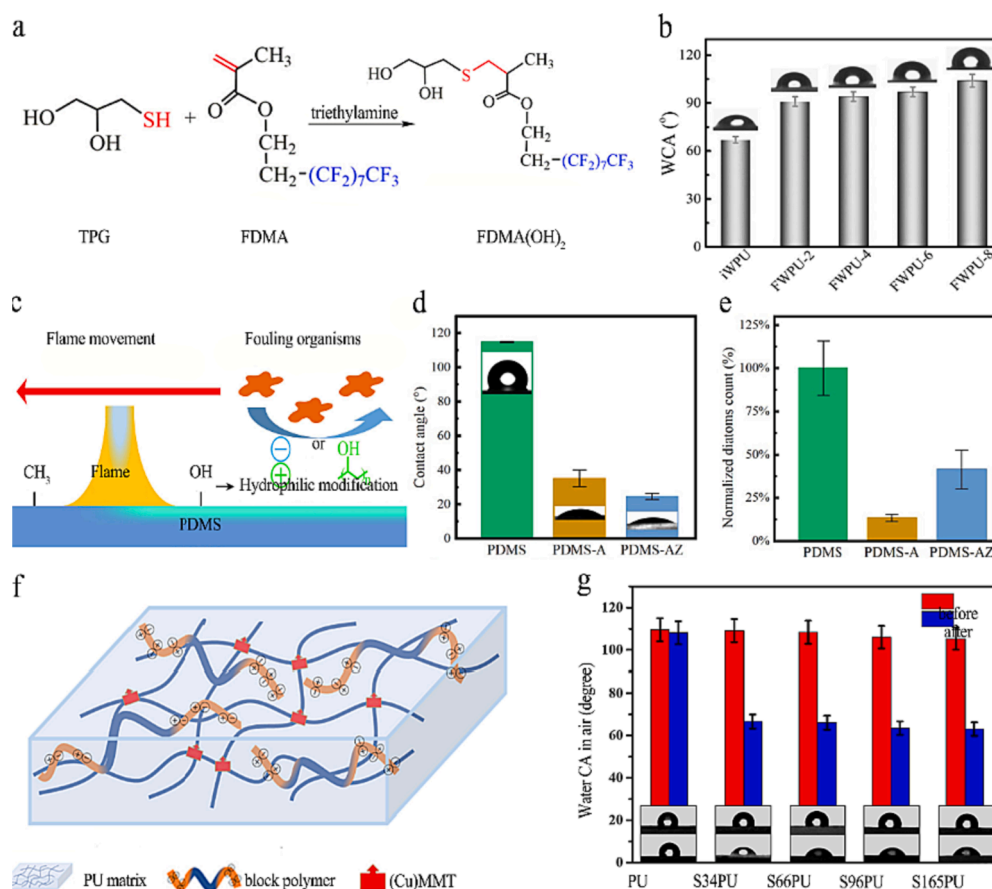


Fig. 4. (a) Synthesis of FDMA(OH)₂ using TPG and FDMA via the thiol-ene 'click' reaction. (b) Comparison of WCA between iWPU and FWPU varying in fluorine composition [60]. Copyright 2021. Adapted with permission from Elsevier. (c) Flame-treated non-toxic polydimethylsiloxane (PDMS). (d) WCA of samples. (e) Quantification of diatom presence in samples after normalization, presented as mean $\pm$ standard error [69]. Copyright 2022. Adapted with permission from Elsevier. (f) Preparation of amphiphilic PDMS-PDMAEMA coating. (g) The WCA of different coatings was evaluated both before and after immersion in water for 24 h [72]. Copyright 2023. Adapted with permission from Elsevier.

exhibited contact angles greater than 90°, which can be attributed to the hydrophobicity of silicone rubber. In the control group, the contact angle of the transparent layer was approximately 102.5°, while the contact angle of the porous layer was about 116.8°. The larger contact angle of the porous layer can be attributed to the presence of a porous structure, which increases the surface roughness [62].

2.3.2. Hydrophilic antifouling mechanisms

The utilization of hydrophilic surfaces has become a prevalent approach to eliminate microbial colonization. The introduction of hydrophilic components induces the formation of hydration layers on the surface, which imparts resistance against biofouling. This resistance is achieved by the creation of a tightly bonded hydration layer that serves as both an energy and physical barrier, effectively impeding various biofouling processes such as protein attachment, initial bacterial attachment, and subsequent microbial membrane formation [63,64]. At the same time, hydrophilicity is an ideal surface modification of polymer materials. By skillfully integrating surface chemistry and roughness properties, it becomes feasible to attain a hydrophilic surface exhibiting a water contact angle that is less than 10°. The extraordinary hydrophilicity of this surface renders it advantageous across a diverse array of research and commercial domains. These encompass water collection, self-purification, prevention of biological fouling, reduction of glare, augmentation of heat transfer during boiling, as well as biomedical applications [65]. Currently, the primary hydrophilic polymers demonstrating efficacy in deterring the adhesion of marine organisms consist of polyethylene glycol (PEG), acrylic ester, and zwitterionic

polymers [66,67]. Li et al., incorporated polyethylene glycol (PEG) into the soft segment of polyurethane (PU) to create poly(ethylene glycol)-segmented polyurethane (PEG-PU) through a two-step polymerization reaction. As the PEG content increased, the surface free energy decreased from 59.36 to 24.55 mJ/m². The water contact angles for PEG₀, PEG₂₅, and PEG₅₀ were determined to be 66°, 41°, and 8°, respectively [68]. Xu et al reported that sulfobetaine silane was grafted to the flame-treated non-toxic polydimethylsiloxane (PDMS) (Fig. 4c). The contact angle (CA) of the treated PDMS was found to be lower compared to the untreated PDMS. Following a treatment duration of 4 s, the CA decreased from 116.6° to 36.6°. The CA values for the samples treated for 8 s and 12 s were measured to be 47.0° and 37.6°, respectively. The polyvinyl alcohol (PVA)-modified flame-treated PDMS (PDMS-A) displayed a 13.4 % attachment of *N. closterium*, compared to the untreated control PDMS. The hydrophilic PVA and sulfobetaine methacrylate (PSBMA)-modified flame-treated PDMS (PDMS-AZ) displayed a 26.9 % attachment of *N. Closterium* (Fig. 4d and e) [69].

2.3.3. Amphiphilic antifouling coatings

The efficacy of antifouling mechanisms that are exclusively hydrophobic or hydrophilic is restricted. Conversely, amphiphilic antifouling mechanisms, characterized by the presence of both hydrophilic and hydrophobic components, exhibit enhanced effectiveness in preventing marine biofouling due to the synergistic interaction between these components. The mechanisms have gained attention in the past decade and are presently regarded as a promising strategy for achieving targeted interfacial functionality in various industries and technologies

[70]. The presence of hydrophilic moieties in amphiphilic mechanisms can provide both energetic and physical barriers, thus impeding the adsorption of fouling substances like proteins, while hydrophobic moieties exhibit excellent antifouling and fouling-release characteristics [71].

Liu et al., designed a series of amphiphilic block copolymer polydimethylsiloxane-poly(2-(dimethylamino) ethyl methacrylate) (PDMS-PDMAEMA) via atom transfer radical polymerization. The copolymer was introduced to the self-healing polydimethylsiloxane-based polyurethane (PDMS-PU) system (Fig. 4f). After being immersed in water for 24 h, the water contact angles (WCAs) of the coatings showed a significant decrease (Fig. 4g). This decrease can be attributed to the gradual migration of hydrophilic PSBMA segments to the surface of the coating during immersion [72].

Surface wettability mechanisms demonstrate remarkable antifouling characteristics. Nevertheless, there are also some shortcomings. (1) The efficacy of the majority of hydrophobic antifouling mechanisms is dependent on the shear force of seawater flow, rendering them effective solely when the coated object is in motion. Moreover, certain low surface energy substances present in hydrophobic coatings have the potential to leach out and contaminate the surrounding environment. Hence, hydrophobic antifouling mechanisms are not suitable for static marine environments, and their performance may exhibit substantial variability over time. (2) Compared to hydrophobic antifouling mechanisms, hydrophilic antifouling mechanisms exhibit reduced environmental harm. However, the susceptibility of their surface hydrophilic polymers to hydrolysis and oxidation poses a risk of denaturation. Under specific circumstances, the surface with hydrophilic property may attract proteins and decrease their antifouling performance. Furthermore, enhancements in mechanical strength are necessary. (3) Amphiphilic antifouling mechanisms possess a unique combination of hydrophilic and hydrophobic properties, owing to the presence of both hydrophilic and hydrophobic groups. This dual nature makes it unfavorable for proteins or glycoproteins to adhere to a surface functionalized with amphiphilic copolymers. Furthermore, the presence of complex surface structures on amphiphilic surfaces can effectively interfere with the formation mechanisms of biofouling during colonization phases, resulting in improved resistance to fouling organisms. However, the antifouling capabilities in actual marine conditions, the mechanical property and long-term durability of amphiphilic surfaces necessitate further assessment. Furthermore, the substantial cost associated with their utilization in the marine environment should be considered. Despite some limitations, surface wettability mechanisms are still widely recognized as an effective and environmentally friendly antifouling strategies.

2.4. Photocatalytic antifouling coatings

The utilization of photocatalytic bactericidal mechanisms is based on the redox capabilities of semiconductor photocatalysts in the presence of light, thereby inducing the generation of reactive oxygen species (ROS) to decompose or transform biofouling. Compared to traditional antifouling strategies, photocatalysis presents notable advantages such as reusability, long-term stability, and environmental friendliness [73,74].

Currently, researches have primarily concentrated on enhancing the photocatalytic activity and visible light of utilization [75]. The utilization of sunlight is restricted by the low photoefficiency of the bandgap, which accounts for only 3–4 % of the solar spectrum, resulting in the underutilization of 44–47 % of the visible light. Furthermore, the photocatalytic efficiency decreases further due to the rapid recombination of photogenerated charges [75,76]. To overcome these shortcomings of the photocatalytic bactericidal mechanisms, the integration of inorganic metal oxides and other hydrophilic materials emerges as a promising strategy to enhance antifouling performance [77]. Titanium dioxide (TiO_2) is one of the inorganic metal oxides widely used in photocatalytic bactericidal mechanisms. TiO_2 is characterized by its relatively low cost,

eco-friendliness, and chemical inertness, which collectively render it an effective inhibitor against the contact and adsorption of organic pollutants on surfaces. The remarkable photocatalytic and hydrophilic properties of TiO_2 , coupled with its superior hydration performance and steric hindrance membrane structure, synergistically contribute to its efficacy [78,79]. Despite the occurrence of photocatalysis on TiO_2 , the potential applications of this process are limited by the wide band gap energy and high recombination rate of TiO_2 . To address this limitation, various attempts have been made, such as introduction of metal and non-metal dopants, the creation of heterojunction structures, the regulation of reaction conditions, and the enhancement of membrane porosity [75]. Sunida et al., investigated the preparation of composite metal oxides with varying tungsten content through a sol-gel process, using titanium isopropoxide and sodium tungstate dihydrate as precursors. The study aimed to enhance the formation of hydroxy radicals ($\bullet\text{OH}$) and evaluate the photocatalytic activity of the composite metal oxides. The results showed that the composite metal oxides containing 10 % by mole of WO_3 (10 % WO_3/TiO_2) exhibited greater photocatalytic activity. The CFU/mL value of a Gram-negative strain (*Pseudomonas sihuiensis*) and a Gram-positive strain (*Exiguobacterium profundum*) present in seawater decreased under the light treatment of composite metal oxides [80].

In addition to TiO_2 , other metal oxides also have photocatalytic activity. Zinc oxide (ZnO) has emerged as a promising multifunctional inorganic compound material because of its strong photocatalytic activity, non-toxicity, and excellent mechanical properties. For example, Lv et al., utilized chemical corrosion technology to produce ZnO hexagonal nanorings (NR ZnO) by combining zinc oxide nanorings with carbon nitride nanorods (Fig. 5a). The UV spectrum analysis revealed that the absorption peak of *Chlorella* occurred at 686 nm, with the intensity reflecting the *Chlorella* quantity on each sample at different time intervals in the seawater solution (Fig. 5b and c). On the fourth day, the peak at 686 nm nearly vanished, and the UV absorption curve appeared as a straight line, indicating the near-death state of *Chlorella*. This observation demonstrated the remarkable antialgae effect of NR ZnO -CNs [81]. Bi_2WO_6 is known as one of the simplest layer-structured bimetallic oxides as a typical n-type semiconductor with a direct bandgap of about 2.8 eV. It has gained recognition for its excellent visible-light photocatalytic performance and favorable bandgap position. As a result, Bi_2WO_6 has been extensively studied in combination with other semiconductors to create composites and enhance the photocatalytic performance further. Include $\text{TiO}_2/\text{Bi}_2\text{WO}_6$ [82], $\text{Bi}_2\text{MoO}_6/\text{Bi}_2\text{WO}_6$ [83], and $\text{Bi}_2\text{WO}_6/\text{BiVO}_4$ [84]. Xiang et al., utilized a simple chemical etching method to prepare a series of $\text{Bi}_2\text{WO}_6/\text{BiOI}$ composite photocatalysts with varying molar percentages of Bi_2WO_6 . The photocatalytic antifouling activities of these composites under visible light irradiation were evaluated using *P. aeruginosa* (3.5×10^6 cfu/mL) as a representative marine fouling microorganism. In the dark and blank control experiments, the number of bacterial cells remained almost unchanged, indicating the non-toxicity of the photocatalytic material. With the increase of Bi_2WO_6 molar percentage from 10 % to 30 %, the photocatalytic antibacterial properties of the $\text{Bi}_2\text{WO}_6/\text{BiOI}$ composite photocatalysts increase obviously. After 45 min of reaction with 30 % $\text{Bi}_2\text{WO}_6/\text{BiOI}$, *P. aeruginosa* only survived at a level of about 2.6 log, resulting in an antimicrobial rate of over 99.99 %. The photocatalytic antibacterial experiments were conducted using 1 mg/mL 30 % $\text{Bi}_2\text{WO}_6/\text{BiOI}$ against the Gram-negative bacteria *E. coli* (2.0×10^6 cfu/mL) and the Gram-positive bacteria *S. aureus* (2.0×10^6 cfu/mL). The presence of 30 % $\text{Bi}_2\text{WO}_6/\text{BiOI}$ resulted in a 99.99 % antimicrobial activity against both *E. coli* and *S. aureus* within 60 min, demonstrating excellent broad-spectrum antimicrobial properties of the material [85].

The range of active components in photocatalytic antifouling coatings are not limited to metal systems, with increasing interest in conjugated organic materials. Due to the structural designability of conjugated polymers as organic polymers, their molecular composition determines the photophysical properties. Additionally, the

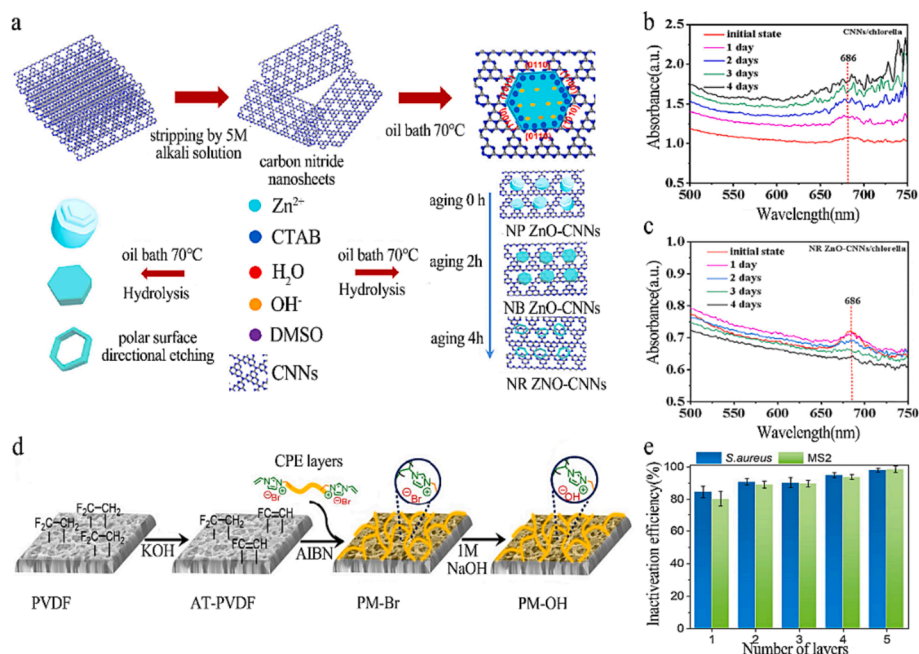


Fig. 5. (a) A diagram showing the process of preparing hybrids consisting of NR ZnO-CNNs. (b) The CNNs/Chlorella over various time periods (1–4 days). (c) The NR ZnO-CNNs/Chlorella over different time intervals (1–4 days) [81]. Copyright 2022. Adapted with permission from Elsevier. (d) A diagram illustrating the synthetic procedure for chemically attaching CPE to a PVDF membrane to produce PMs. (e) Assessment of the effectiveness in deactivating *S. aureus* and MS2 on PM-OHs under visible light for one hour [87]. Copyright 2022. Adapted with permission from Elsevier.

characteristics of conjugated polymers, such as hydrophilicity, hydrophobicity, machinability, and potential reactivity can be adjusted through side-chain engineering [86]. Among them, chemical graft polymerization is widely used for the modification of conjugated polymers because of its simple technique and better regulation of the degree of functionalization. Jeong et al., successfully demonstrated the chemical grafting of conjugated polyelectrolytes (CPE) onto commercially available polyvinylidene fluoride (PVDF) membranes. This process resulted in the creation of hydrophilic photocatalytic membranes (PMs) that possess an antibiofouling property (Fig. 5d). The PMs exhibited remarkable antimicrobial activity against both *S. aureus* and MS2 bacteriophage. After 1 h of visible light irradiation, the PMs achieved an impressive inactivation efficiency of 98.1 % for *S. aureus* and 98.6 % for MS2 bacteriophage (Fig. 5e) [87].

Photocatalytic bactericidal mechanisms present unique advantages and are considered as one of the options for marine biofouling control. However, prior to their practical application, numerous challenges necessitate attention. The performance of photocatalytic bactericidal mechanisms can be affected by multiple factors, such as the temperature, pH level of the water solution, dissolved oxygen, and intensity of light. In addition to these external variables, it is essential to understand the internal mechanism of photocatalytic processes in order to enhance its antifouling efficiency. This encompasses the investigation of the photocatalyst's function in the antifouling process and the analysis of fouling organisms' response at the biomolecular scale. The exploration of these antifouling mechanisms will provide valuable insights for the design of photocatalysts for large-scale industrial applications. Therefore, additional investigation is imperative to formulate novel photocatalytic materials with broad-spectrum and high efficiency, understand theoretical data and deactivation mechanisms.

2.5. Biomimetic antifouling mechanisms

The term biomimetic refers to imitating or being inspired by nature in the process of designing materials or structures. Bionics encompasses three dimensions: chemical and molecular bionics, biological mechanism bionics, and physical structure bionics. The current approach to

biomimetic antifouling mechanism is based on two strategies: the utilization of natural antifouling agents (Section 2.1.1), and the development of biomimetic surface micro-morphology. These strategies aim to develop novel and environmentally friendly antifouling technologies.

After a long period of natural evolution, numerous organisms in the natural world show an excellent antifouling ability, which is related to the development of distinct microscopic and nano-structured surfaces through evolution. In contrast to artificial structures, the skin of various marine organisms (e.g. sharks and whales) experiences significantly less biofouling [88]. This inherent antiadhesion characteristic has served as the foundation for the development of biomimetic antifouling mechanisms, with bionic shark skin being particularly noteworthy. Chien et al., investigated the microscale structure of shark skin denticles at abdomen (A) and fin (F) locations (as depicted in Fig. 6a). The researchers analyzed the roughness and wetting properties associated with these microstructures and also assessed the impact of surface properties on early bacterial attachment and biofilm formation. The findings indicated that the rough patterns on shark skin facilitated bacterial attachment during the initial stages. However, these patterned surfaces restricted the growth of biofilm. Although there was a slight increase in absorbance values after 14 days, the OD values remained consistently low (0.51 on A2 and 0.47 on F1) (Fig. 6b) [89]. Lin et al., replicated the bionic (shark skin) structure onto the outer layer of the polydimethylsiloxane (PDMS) membrane to create a hydrophobic surface. As depicted in Fig. 6c, the bionic outer layer PDMS membranes exhibited a higher contact angle of 120°, compared to the pristine PDMS with a contact angle of 95°, indicating successful replication of the shark skin structure. Fig. 6d demonstrated that WCA of PDMS/PVA-PMPC (grafting PMPC (2-methacryloyloxyethyl phosphorylcholine) onto the PDMS-PVA (polyvinyl alcohol) membrane) was less than 10° while the water was completely absorbed within 3 min, showing the super-hydrophilicity [90].

Lotus leaf reproduction has recently emerged as a biomimetic antifouling technology. The lotus leaf exhibits a high contact angle exceeding 150° and a natural ability to repel water droplets, which imparts hydrophobic properties. Moreover, the lotus leaf utilizes its high contact angle to facilitate the sliding of droplets at a minimal angle of

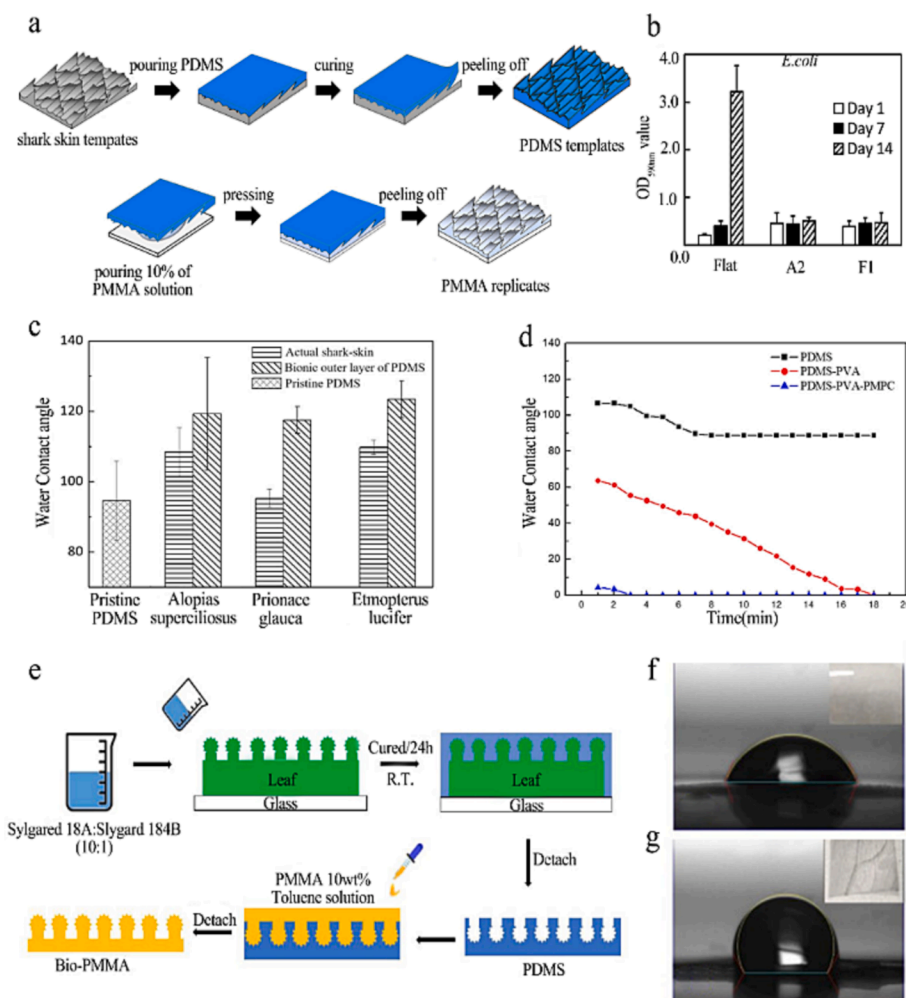


Fig. 6. (a) The replication process for the surface of a shark's skin. (b) The quantification of residual *E. coli* biofilms was done using crystal-violet staining [89]. Copyright 2020. Adapted with permission from Elsevier. (c) The measurement of water contact angle (WCA) was conducted on three different samples: the original PDMS membrane, the realistic artificial shark skin structures, and the PDMS membranes with bionic outer layers. (d) Over time, the changes in the WCA tendency were investigated within the internal layer of PDMS, PDMS-PVA, and PDMS-PVA-PMPC membranes [90]. Copyright 2020. Adapted with permission from Elsevier. (e) A negative soft template, silicone rubber (PDMS), was prepared using a nano-casting technique to transfer the pattern of natural XSL onto artificial PMMA coatings (Bio-PMMA). (f) The water droplet contact angle of PMMA was examined. (g) Water droplet contact angle of Bio-PMMA [93]. Copyright 2023. Adapted with permission from Elsevier.

less than 5°, causing them to effortlessly roll off the leaf surface. Inspired by this phenomenon, the lotus leaf-inspired mechanism exhibits exceptional hydrophobicity (water-repellent) and self-cleaning ability [91]. Consequently, this innovative concept offers a promising avenue for investigating novel bionic antifouling technologies. For example, Yang et al., prepared three hierarchical superhydrophobic copper coatings with different microstructure sizes similar to the lotus leaf by facile one-step pulse electrodeposition. The self-cleaning performances showed coatings are wholly cleaned in 15 s with ~ 20 water droplets, which exhibits excellent hydrophobic and antifouling properties [92].

The application of nano-casting technology has been employed to replicate the surface structure of *Xanthosoma sagittifolium* leaf (XSL), leading to the creation of bio-nanomaterials exhibiting superhydrophobicity and high surface area characteristics. Furthermore, these materials have also demonstrated excellent performance in preventing bacterial adhesion and resisting biofilm formation. Hwang et al., fabricated a hydrophobic surface coating of Poly (methyl methacrylate) (PMMA) using the nano-casting technique. The process involved transferring the surface from a natural leaf of *Xanthosoma sagittifolium* onto a polymer surface (Fig. 6e). To assess the hydrophilic properties of the surface, the static water contact angle (CA) on the PMMA surface was

measured and found to be $68^\circ \pm 0.8^\circ$ (Fig. 6f). The CA results indicated that the flat PMMA surface became hydrophilic. For the XSL replicas, the CA value on surfaces was determined to be $121^\circ \pm 0.6^\circ$ (Fig. 6g). The findings demonstrated that the inherent hydrophilic characteristic of PMMA can undergo transformation into hydrophobicity upon replication of the XSL template. This evidence substantiated the notion that such microstructures exhibited the capability to augment the hydrophobic nature of the material's surface [93].

In conclusion, biomimetic antifouling mechanisms demonstrate surface micromorphology, fouling release capability, and the potential for the 'Lotus effect', making them promising candidates for marine antifouling applications. Nevertheless, their mechanisms have garnered less attention and remain unclear. Additionally, certain researchers express reservations regarding the antifouling effect of biomimetic micromorphic antifouling mechanisms, citing the colonization of fouling organisms on the skin of specific marine organisms. The primary emphasis in recent research has been on the replication of surface microstructures; however, these approaches encounter notable obstacles including high costs and complications in large-scale production and application.

As the awareness of environmental protection continues to grow,

there is still much work to be done in developing long-term effective, simple, safe, and environmentally friendly anti-fouling materials or technology. Firstly, it is important to focus on the development of marine anti-fouling materials that are environmentally friendly and non-toxic. This requires in-depth research into the potential ecological toxicity of antifouling materials and the elimination of harmful ones. Secondly, improving the mechanical properties of anti-fouling materials, such as tensile force and adsorption force, is crucial for enhancing their durability. Additionally, incorporating functional groups with different anti-fouling effects through block copolymerization can effectively enhance the overall anti-fouling performance of the material. It is important to note that relying on a single anti-fouling method is not sufficient to achieve the desired effect in practical applications. Therefore, future research should focus on integrating multiple methods to develop multi-mechanism synergistic anti-fouling materials.

3. Antifouling systems for seawater resource utilization

The ocean contains a substantial array of resources, encompassing fuel, biological, chemical, and other resources. In the process of developing and utilizing seawater resources, emphasis is frequently placed on how to efficiently extract resources based on the structural characteristics of the materials themselves, while ignoring the influence of the marine environment. It is widely acknowledged that any object that encounters seawater faces the challenge of marine biofouling. Hence, when designing materials for seawater resource utilization, it is imperative to take into account their structural attributes and antifouling properties. Based on the above marine antifouling strategies (Section 2), the biofouling control in extraction of uranium and lithium from seawater, as well as net cage for aquaculture is emphasized.

Uranium (U), a vital primary resource for nuclear power, is estimated to be present in the ocean in quantities of approximately 4.5 billion tons. These resources are widely acknowledged as the prospective foundation for the advancement of nuclear power. Consequently, the development of a cost-effective and efficient technology for the extraction and enrichment of uranium from these resources in a sustainable manner assumes paramount importance in guaranteeing the robust growth of nuclear power. Compared to terrestrial uranium mining, the process of extracting and enriching uranium from seawater is deemed to be a more economically viable and environmentally sustainable approach. The substantial uranium resources can be acquired while causing minimal harm to the surrounding environment. As a result, uranium extraction from seawater is regarded as a promising and efficient endeavor in the development of nuclear fuel resources [99–101]. The ocean is a complex ecosystem that sustains a diverse community of life, including organic matter, microorganisms, marine plants, and animals. These biological systems exert a substantial influence on the efficacy of adsorbents employed in seawater uranium extraction. Biofouling, a phenomenon resulting from the accumulation of micro- and macroorganisms or their excretion products on surfaces, can lead to a significant decline of up to 30 % in the performance of seawater uranium extraction. This is attributable to the detrimental effects on the adsorbent, thereby its capacity for reusability. Therefore, the development of uranium extraction adsorbents with antifouling performance is imperative in order to optimize efficiency and minimize extraction costs. The attainment of this objective will also promote the practical application of seawater uranium extraction technology. Extensive research conducted by domestic and international researchers has resulted in significant advancements in the fundamental principles, extraction mechanism, and extraction process of seawater uranium extraction technology.

Lithium (Li) holds significant importance in contemporary society owing to its extensive utilization across diverse sectors encompassing ceramics, glass, pharmaceuticals, nuclear industries, and battery technologies. Lithium-ion batteries, renowned for their exceptional energy storage capabilities, are experiencing a surge in popularity due to their minimal self-discharge, absence of heavy metal pollution, lack of

memory effect, high energy density, prolonged cycle life, and eco-friendliness. Consequently, the demand for lithium has witnessed a notable upswing. Seawater exhibits significant potential as a lithium source, boasting a lithium concentration that surpasses that of ore and brine sources by a factor of 5000 [102]. The primary techniques employed for extracting lithium from seawater involve membrane separation and adsorption. However, both methods are equally susceptible to fouling organisms and reduced interaction between substrate molecules and the adsorbent or separation membrane. Consequently, these factors result in decreased efficiency of lithium extraction from seawater.

In recent years, aquaculture technology has been a focus of marine biological resource development. The culture net box, due to its substantial specific surface area, offers an optimal environment for the colonization of fouling organisms. Additionally, the net box environment is abundant in residual bait, feces, secretions, and other nutrients, which significantly enhance the attachment and growth of fouling organisms on the culture facilities. The increase in fouling organisms adds extra weight to the culture facilities, leading to the distortion and deformation of the nets and, occasionally causing damage. As a result, the application of antifouling systems as a means to alleviate the adverse effects of marine fouling organisms on the use of seawater resources has emerged as a novel direction. This advancement highlights the need for effective antifouling materials in aquaculture technology.

3.1. Uranium extraction from seawater

Adsorption is commonly regarded as a promising approach for the extraction of uranium from seawater due to its operational simplicity, low energy consumption, high enrichment capacity, and rapid separation rate. Generally, the sorbent possesses the following features. (1) In terms of macro/microstructure, the sorbent necessitates robust mechanical strength to efficiently collect and endure the intricate conditions of the natural marine environment. (2) A porous three-dimensional structure is essential to increase the contact area between the uranium and the sorbent. (3) Functional ligands need to exhibit a high affinity to effectively capture uranium. (4) Antifouling measure is essential for marine sorbents to mitigate the obstruction of their active sites by microorganisms [103,104]. Extensive research has been conducted on antifouling materials for uranium extraction from seawater, including inorganic particulate adsorbent materials, polymer-based adsorbent materials, and natural biomass adsorbent materials. Herein, the corresponding antifouling characteristic is focused in this section.

3.1.1. Inorganic particulate adsorbent materials

Metal oxide nanoparticles are utilized as antimicrobial agents due to their low toxicity, heat-resistance, and effectiveness against microbial pathogens. Currently, the most commonly used inorganic antimicrobial agents include Ag, TiO₂, Zn series, and other inorganic nano antimicrobial agents.

Silver has long been recognized as an excellent antimicrobial material owing to its efficient, broad-spectrum antimicrobial ability [105–107]. Guo et al., synthesized a composite material consisting of graphene oxide, silver ions, and zeolitic imidazolate framework-67 (Fig. 7a). The concentration of uranium ions in seawater was determined to be $3.394 \pm 0.143 \mu\text{g L}^{-1}$. Fig. 7b revealed that the zeolitic imidazolate framework-67 (GZA) composites exhibited a high adsorption amount ($4.974 \mu\text{g g}^{-1}$) and removal rate (73.27 %) when subjected to real seawater. Additionally, Fig. 7c demonstrated that there was no significant change in the removal of U(VI) by GZA composites at different initial concentrations ranging from 16.04 to $227.23 \mu\text{g L}^{-1}$ [105]. In addition, silver nanoparticles exhibit superior antifouling properties compared to conventional silver ionic materials due to their size effect and surface effect, etc. For example, Mo et al., developed a novel core-shell nanoparticle containing amidoxime groups on its surface. This nanoparticle can effectively bind uranyl ions in seawater,

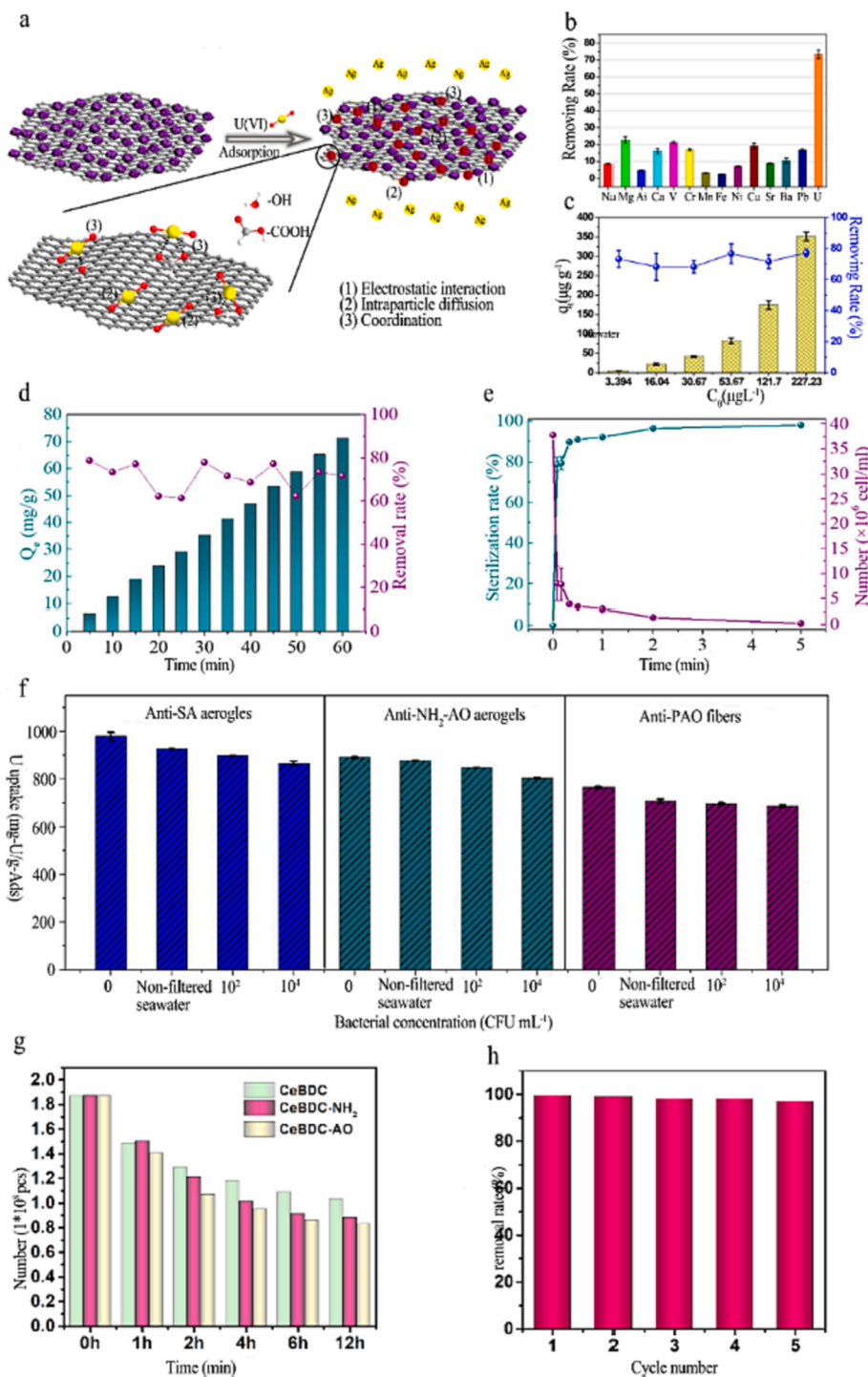


Fig. 7. (a) The proposed mechanism of U(VI) adsorption on GZA composites. (b) The adsorption selectivity of GZA composites for U(VI). (c) The adsorption capacities and removal rate of GZA composites for U(VI) in triplicate seawater samples [105]. Copyright 2020. Adapted with permission from Sustainable Energy & Fuels [109]. Copyright 2022. Adapted with permission from Elsevier. (d) The extraction ability of TiO_2/FTO for U under UV light irradiation for a brief duration. (e) The efficacy of UV irradiation in water for bacterial eradication [109]. Copyright 2022. Adapted with permission from Elsevier. (f) Uranium adsorption capacities in the artificially prepared seawater containing bacteria [112]. Copyright 2020. Adapted with permission from Elsevier. (g) A comparison of three materials by algae removal performance (the number of algae in the initial solution was 1.87×10^8). (h) The adsorption recyclability of CeBDC-AO after five cycles is examined using a uranium solution with an initial concentration of 100 mg L^{-1} [121]. Copyright 2020. Adapted with permission from Elsevier.

while simultaneously releasing silver ions that effectively inhibited the proliferation of *E. coli* and *Staphylococcus aureus* at high concentrations (removal rate of 99 % at 50 mg/L). The adsorption capacity of this adsorbent in aqueous solution was determined to be 234.74 mg/g [108].

In recent years, TiO_2 has attracted significant attention due to its antibacterial properties, affordability, self-cleaning capabilities,

favorable hydrophilicity, and stability. When exposed to ultraviolet light and water vapor, TiO_2 can be activated to generate oxide anions and hydroxyl radicals. These highly reactive oxygen-containing radicals exhibit efficient oxidation of organic cells, ultimately resulting in the death of microorganisms. Liu et al., prepared TiO_2 films on the surface of fluorine-doped tin oxide (FTO) by magnetron sputtering. As depicted in

Fig. 7d, the continuous rate of uranium extraction from TiO_2/FTO was observed to exceed 70 % within a duration of 5.0 min when subjected to UV irradiation. Moreover, the cumulative amount of uranium extracted reached 71.13 mg/g after 60 min (with initial concentration = 50 $\mu\text{g/L}$, volume of 200 mL, and pH = 6.0). Fig. 7e demonstrated that the sterilization efficiency of the UV lamp was measured at 90.85 % within a duration of 30 s, and the concentration of cells per unit volume in the solution remained below 5×10^6 . After exposing the solution to UV illumination for a duration of 5.0 min, the sterilization efficiency increased to 99.39 % [109].

3.1.2. Polymer-based adsorbent materials

Polymer-based adsorbent materials with abundant functional groups demonstrate superior antifouling activity compared to conventional adsorbents. Polymer-based adsorbent materials present enhanced durability and reusability in the uranium extraction from seawater environments. Typically, these materials comprise a matrix material and a ligand group, which can be grafted onto the matrix material to create a highly effective uranium adsorbent. In the design of such materials, antibiofouling functional groups, including amino, quaternary amine, and guanidine groups, are commonly employed [110,111]. Moreover, the stability of the polymer in the long-term marine environment is attributed to its abundant covalent bonding and stable cross-linking of the polymer groups. The introduction of the specific functional groups further enhances the adsorption capacity of the polymer to uranium, thus improving its overall effectiveness.

For example, Shi et al., developed polyaminoxime nanofibre membranes using the bacterial cellulose-assisted ionic cross-linking method. These membranes exhibited high-strength and biofouling-resistant properties, making them well-suited for uranium recovery. The adsorption capacity of the membranes was determined to be $1013 \pm 9.7 \text{ mg-U/g-Ads}$ in an 8 ppm uranium spiked solution. Additionally, the adsorption amounts in seawater containing bacteria ranged from 90.67 % to 94.82 % of those in sterilized seawater, indicating the beneficial antifouling effects of the prepared membranes for uranium extraction (Fig. 7f) [112].

3.1.3. Natural biomass adsorbent materials

Biomass is a renewable organic matter that holds potential as a valuable material for uranium adsorbents. This approach not only aligns with environmentally friendliness but also proves economically advantageous. Biomass materials offer numerous benefits, including wide availability, abundant reserves, affordability, and minimal concerns regarding secondary pollution. Moreover, biomass frequently encompasses active functional groups like carboxyl, amino, and hydroxyl, which facilitate uranium adsorption and chemical modification. However, it is important to acknowledge certain limitations associated with a single type of biomass: (1) the adsorption capacity is limited due to the relatively homogeneous nature of biomass; (2) certain biomaterials pose difficulties in the separation of adsorbed uranium, leading to high costs associated with separation; (3) improving the physical and mechanical properties of biomass is necessary for its effective use in industrial applications [113].

For example, Bai et al., developed a unique 3D reticular antifouling green adsorption material by covalently grafting polyethylenimine (PEI) and guanidineacetic acid (GDAC) onto the surface of hemp fibers (HF). In the scenario where U (VI) concentration was low (ranging from 3.05 to 99.03 $\mu\text{g/L}$), the adsorbent exhibited a consistently high removal rate (over 85 %). Moreover, the adsorbent maintained a substantial adsorption efficiency (above 72 %) even in solutions of U (VI) containing algae at various temperatures [114]. Yuan et al. conducted a study on *Bacillus velezensis* strain UUS-1, which revealed its exceptional attributes including rapid and high capacity for uranium uptake, strong tolerance to both salt and uranium, and extensive antimicrobial activity through the generation of antimicrobial metabolites. The equilibrium adsorption capacities of the strain in simulated seawater with uranium

concentrations of 10, 20, 50, and 100 ppm were measured at 285.64 ± 14.73 , 335.13 ± 17.18 , 406.25 ± 24.63 , and $483.31 \pm 31.34 \text{ mg U g}^{-1}$ dry bacterial cell debris, respectively. Additionally, when seawater was cultivated in a medium enriched with UUS-1's fermentation broth and supplemented with 2 % of the broth, a noticeable hindrance in the growth of marine microorganisms was observed. After a 6-hour co-cultivation period, the bacterial concentrations in the cultures treated with UUS-1's fermentation broth and those untreated were determined to be 1.59×10^4 and $1.60 \times 10^6 \text{ CFU mL}^{-1}$, respectively [115].

Chitosan (CS), a naturally occurring biomass possessing positive charges, demonstrates a promising antimicrobial impact on a wide range of microorganisms, including bacteria, yeast, fungi, and other similar entities. The encounter of CS's positive charge with the negatively charged membranes of microbial cells triggers the liberation of bacterial proteins along with various cellular constituents, thereby producing an antimicrobial effect [116–118]. Li et al., developed a novel environmentally-friendly anti-hydrogel by modifying chitosan with glyphosate (bPG). The amount of uranium adsorbed at equilibrium increased as the water temperature rose, implying that the adsorption process of the anti-hydrogel was endothermic. This observation aligns with the thermodynamic properties exhibited by other phosphate-based adsorbents. Following exposure to simulated sunlight irradiation for 85 h in 8 ppm U-spiked water, the anti-hydrogel demonstrated a saturated uranium adsorption capacity of 253 mg-g^{-1} . This capacity was 54 % higher than that of the control group which remained unexposed to light. Moreover, the anti-hydrogel exhibited inhibitory effects of 99.9 %, 99.9 %, and 99.2 % against *S. Aureus*, *E. Coli*, and *P. Aeruginosa*, respectively [119].

3.1.4. Porous frame adsorbent material

The extraction of uranium from seawater is constrained by different factors linked to the design of materials, like binding sites that are selective towards U, their uniform distribution, exposure to surface area, diffusivity, and stability. In recent years, porous frame materials have gained significant attention in uranium extraction research due to their larger specific surface area, higher porosity and robust stability.

Metal-organic frameworks (MOFs) are synthesized through the establishment of coordination bonds between metal clusters and organic linkers. These MOFs demonstrate antibacterial activity and present advantages over metal or metal oxide nanoparticles. MOFs are particularly remarkable owing to their high specific surface area and promising adsorption properties, which make them suitable for potential applications in the uranium extraction from seawater [120]. Chen et al., exploited the self-assembly property of MOFs to synthesize cerium terephthalate with shuttle superstructures (CeBDCs) in a short period of time at room temperature by using the effective coordination between Ce^{3+} and terephthalic acid (1–4-benzenedicarboxylic acid). In the solution with a concentration of algae at $1.87 \times 10^8 \text{ pcs/mL}$, these substances demonstrated effective elimination of algae. Among them, CeBDC-AO, which was amidoximated, exhibited an algal removal efficiency of approximately $1.04 \times 10^7 \text{ cells/mg}$. CeBDC and CeBDC- NH_2 , which was aminated, also demonstrated satisfactory ability to remove algae (Fig. 7g). Additionally, Fig. 7h reveals that CeBDC-AO maintained robust adsorption capabilities even after undergoing five adsorption experiments, resulting in a high uranium removal rate of 98 % [121].

Covalent organic frameworks (COFs) possessing precisely pre-designed structures provide functionally specific structural control, setting them apart from conventional porous adsorbents. These COFs exhibit distinctive advantages, including low density, high porosity, excellent stability, structural diversity, ease of functionalization, and controllable pores. Consequently, they have found widespread application in the extraction of uranium from seawater [122]. Chen et al., engineered the pore structure of a multicomponent COF to contain amidoxime groups, triazine groups, and bipyridine-metal groups (COF 4-Pd-AO). Incorporating the functional moieties onto the framework

linkers enabled the attainment of this capability. COF 4-Pd-AO demonstrated exceptional efficiency in extracting uranium, with a daily calculated uptake of 4.62 mg/g from natural seawater under the influence of visible light. COF 4-Pd-AO inhibited *Chlorella elongata* and *Chlorella solani* by 66.48 % and 57.65 %, respectively, under dark conditions [123].

Despite the demonstrated efficacy of anti-biofouling adsorbent materials in suppressing microorganisms growth in seawater, certain limitations persist. (1) A notable drawback is the inherent instability of the non-covalent bonding between the inorganic antimicrobial agent and the adsorbent material, which makes it easy for the antimicrobial agent to detach from the adsorbent. Additionally, the degradation of inorganic nanoparticles poses a formidable challenge, potentially engendering adverse consequences for marine ecology. (2) Polymer-based antifouling adsorbent materials have emerged as the dominant choice for uranium extraction from seawater, due to their high tensile strength, chemical stability, and capacity for large-scale production. However, there remains potential for enhancing the uranium adsorption efficacy of these materials. The additional investigation is imperative to assess their antifouling ability against marine organisms in real marine environments. (3) Biomass antimicrobial agents demonstrate remarkable antimicrobial activity and exhibit compatibility with uranium extraction materials from seawater. Additionally, biomass materials are abundantly available, economically viable, biodegradable, and environmentally friendly, thus promoting sustainable development and low-carbon strategies. Due to their versatile synthetic methods and widespread availability, biomass uranium extraction materials hold significant promise as a viable solution in the field. (4) Porous framework adsorbent materials are commonly employed for extracting uranium from seawater due to their exceptional performance. MOF materials, composed of various metal oxides, provide a diverse array of choices for adsorbing U (VI) in different applications. Furthermore, metal-free frameworks can prevent secondary contamination in real-world environmental scenarios. Certain COFs maintain their crystal structures even when exposed to strong acidic environments for extended periods of time. However, there are several challenges that necessitate further

attention. One significant challenge is the recovery of porous framework materials, typically found in a powdered state, thereby limiting their practical application. Therefore, current research is focused on incorporating porous skeletal materials (e.g., MOF and COF) into macroscopic structures like fiber materials, aerogels, hydrogels, etc.

3.2. Lithium extraction from seawater

The ocean contains low levels of Li^+ (0.2 ppm) alongside high concentrations of competing ions such as Mg^{2+} (40 ppm), which presents challenges for the Li recovery from seawater. To overcome these challenges, various separation techniques including membrane, adsorbent, solvent extraction [124], electrodialysis [125], nanofiltration [126], and precipitation [127].

Among these techniques, membrane separation technology is considered an ideal alternative due to its high porosity, large surface area, and cost-effectiveness. Qiu et al., prepared a polyvinyl alcohol (PVA) reinforced cellulose acetate membrane- $\text{Li}_4\text{Mn}_5\text{O}_{12}$ (PVA/CAM- $\text{Li}_4\text{Mn}_5\text{O}_{12}$) to efficiently extract Li^+ . According to the results presented in Fig. 8a, PVA/CAM- $\text{H}_4\text{Mn}_5\text{O}_{12}$ demonstrated a remarkable Li^+ extraction efficiency of 98.8 % from the simulated seawater. Notably, this efficiency was found to be substantially higher compared to other ions (with the highest separation efficiency among Na^+ , Sr^{2+} , Ca^{2+} , K^+ , and Mg^{2+} being less than 5 %). Moving to Fig. 8b, it can be observed that as the number of separations involving Bovine albumin (BSA) increased, the flux of PVA/CAM- $\text{H}_4\text{Mn}_5\text{O}_{12}$ to pure water observed a decline from 22.22 ($\text{L}/(\text{m}^2\cdot\text{min})$) to 11.99 ($\text{L}/(\text{m}^2\cdot\text{min})$). Referring to Fig. 8c, the corresponding Flux Recovery Ratio (FRR), which measures the percentage of pure water flux before and after the BSA separation, displayed a decrease from 94.28 % to 53.96 %. However, even after four BSA separations, the FRR remained relatively high at 82.34 % [128].

In addition to membrane separation technology, the adsorption method is regarded as a highly promising approach due to its simple operation, high recovery efficiency, and relatively low energy consumption. Ion sieve oxides, classified as adsorbents, have efficient ion recognition and screening properties for the extraction of specific metal

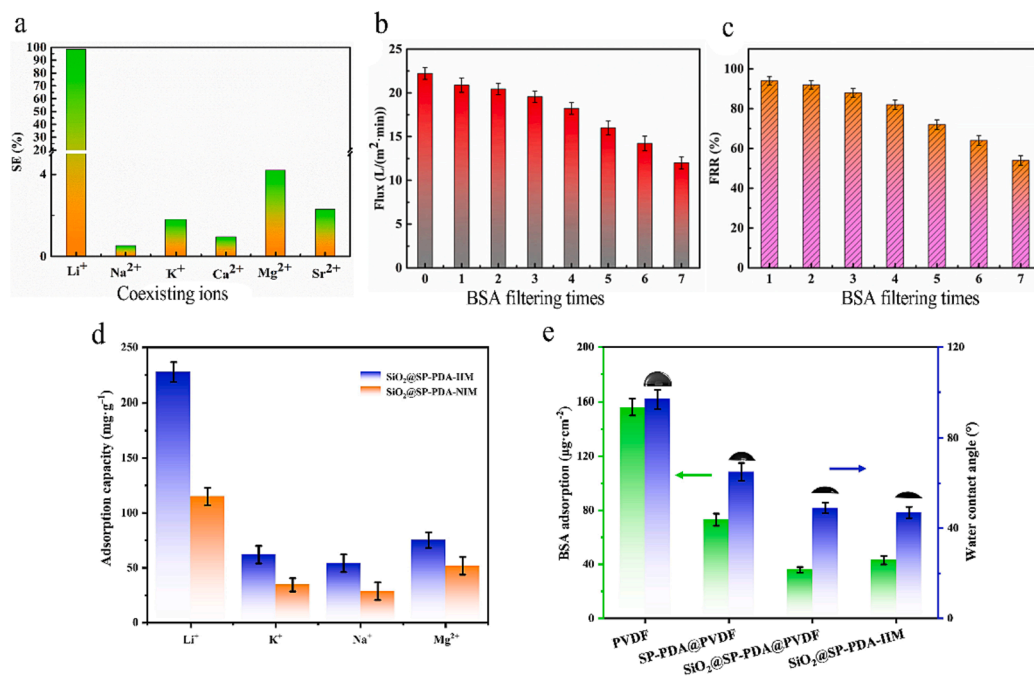


Fig. 8. (a) The selective adsorption of Li^+ under complex conditions (at a temperature of 298.15 K and pH of 8.4). (b) The flux of PVA/CAM- $\text{H}_4\text{Mn}_5\text{O}_{12}$ into pure water following various durations of BSA separation. (c) Antifouling performances of PVA/CAM- $\text{H}_4\text{Mn}_5\text{O}_{12}$ [128]. Copyright 2021. Adapted with permission from Elsevier. (d) The selective adsorption performance of $\text{SiO}_2@\text{SP-PDA-IIM}$ and $\text{SiO}_2@\text{SP-PDA-NIM}$ for various ions was investigated. (e) The adsorption property of BSA onto the prepared membranes was studied under dynamic conditions, along with the water contact angle (WCA) of the fabricated membranes [130].

ions. Lithium ion screens (LIS), which encompass a series of lithium-selective adsorbents with unique chemical structures, demonstrate exceptional proficiency in the separation of lithium ions from seawater. Sun et al., fabricated ion-imprinted antifouling nanocomposite membrane (LiI-NcMs) blended with graphene oxide (GO) and TiO_2 nanomaterial for selective adsorption of lithium ion. Remarkably enhanced hydrophilicity was successfully achieved on LiI-NcMs, as evidenced by a contact angle of 55.1° . LiI-NcMs demonstrated superior stability, as demonstrated by their ability to maintain a maximum adsorption capacity of 88.1 % even after undergoing 20 cycles of adsorption and desorption. The modified membranes, LiI-NcMs, displayed a low BSA adsorption capacity (about $17 \times 10^{-3} \text{ mg} \cdot \text{cm}^{-2}$), indicating a substantial enhancement in their antifouling properties [129]. A novel multilayered interlayer ($\text{SiO}_2/\text{SP-PDA}$) was developed by He et al., for the selective adsorption of lithium ions. To enhance the uniformity of $\text{SiO}_2/\text{SP-PDA}$ modification and minimize ion-imprinted layer agglomeration, polydopamine (PDA) prepared using sodium periodate (SP-PDA) was employed. The modification yielded an increased adsorption capacity of $231.77 \text{ mg} \cdot \text{g}^{-1}$, as shown in Fig. 8d. When subjected to dynamic conditions, the untreated polyvinylidene fluoride (PVDF) membrane exhibited a significantly higher albumin from BSA adsorption capacity ($156.02 \mu\text{g} \cdot \text{cm}^{-2}$) in comparison to the other membranes. This finding suggests that additional modification enhances the membranes' antifouling behavior, as depicted in Fig. 8e [130].

Considering the advantages and disadvantages of various lithium extraction methods, the adsorbent method emerges as a more promising approach, with particular emphasis on the potential of employing ion sieve metal oxide. Nonetheless, challenges persist in both domestic and international contexts regarding lithium extraction. On the one hand, the actual adsorption capacity of the ion sieve remains significantly below its theoretical capacity; on the other hand, the synthesis of more efficient adsorbents has predominantly resulted in powder forms, which is not beneficial for practical applications. Therefore, future research should prioritize the investigation of mechanism of the ion sieve effect, enhancement of the actual adsorption capacity of powder adsorbent, and resolution of issues of high dissolution loss rate, short lifetime, and slow exchange rate of adsorbent. Meanwhile, the synthesis of adsorbent with high lithium adsorption capacity, favorable selectivity, and cost-effectiveness will become the focus of current work and is also the future development direction of lithium extraction by adsorption technology.

3.3. Net cage for aquaculture

Aquaculture has emerged as a significant industry for utilization of marine resources in contemporary times. In light of rapid population growth and the surging demand for aquaculture products, aquaculture faces the dual challenge of ensuring food security while simultaneously adopting sustainable practices [131]. Biological fouling poses a persistent issue across all marine industries, including aquaculture. The implementation of net cage for aquaculture presents a viable approach for the gradual mechanization, modernization, and industrialization of traditional fish farming technology, thereby improving production efficiency. Net cages possess characteristics such as non-toxicity, porosity, and a large surface area, making them easy to attach to fouling organisms due to prolonged underwater exposure and direct contact with the aquatic environment. Additionally, the seawater used for culture contains abundant nutrient salts and aquaculture wastes, which provides sufficient nutrients for the growth of fouling organisms. Failure to promptly remove a substantial population of fouling organisms can cause significant harm to the production of aquaculture [132].

Biofouling exerts various impacts on aquaculture operations, encompassing both infrastructure and farmed organisms [133]. Firstly, it increases the weight of nets and culture equipment, which leads to higher costs for providing buoyancy and anchoring. Additionally, fouling organisms obstruct the nets, limiting water exchange and

reducing oxygen levels. Moreover, the weight of cultured aquatic products, like kelp, is also increased by biofouling, making them more susceptible to breakage and displacement. Fouling organisms further present a direct competition for vital resources, including food, oxygen, and other resources, with farmed products. Furthermore, certain fouling organisms carry disease-causing viruses, bacteria, and parasitic organisms, thereby potentially causing infections and mortality among farmed products [134].

The initial approach employed to address the issue of fouling on net cages involves the manual removal method. This method exhibits several characteristics: (1) It requires low technical prerequisites and can be applied across various situations. (2) However, it suffers from low labor efficiency and substantial labor expenses. (3) Its operational scope is constrained, typically not surpassing a water depth of 20 m. (4) The detachment of fouling organisms during the process may result in environmental pollution. With the expansion of net cage for aquaculture, there is a growing need to find alternative anti-fouling technologies. Drawing inspiration from antifouling technology employed in ships, researchers have developed antifouling coatings specifically designed for nets. For example, Yu et al., modified Cu nanoparticles (CuNPs) using an organic surfactant. Subsequently, CuNP-modified high-density polyethylene (CuNP/HDPE) monofilaments were created through melt spinning. To prepare netting with a mesh length of 25 mm, CuNP/HDPE-1.0 nanocomposite monofilaments with optimal overall performance were twisted into a $36\text{tex} \times 10 \times 3$ netting twine and tested in the shallow sea area of the South China Sea. The CuNP/HDPE nanocomposite mesh exhibited a lower weight gain (reduced by 32 %) compared to the HDPE mesh after a duration of six months [135]. Qiu et al., developed a durable and adaptable composite by utilizing 3D spined fabrics as the core material and polydimethylsiloxane (PDMS) as the shell material. This unique combination enabled the shell component to penetrate the core material, enhancing interlocking mechanisms on both micro and macro scales. As a result, the composite demonstrated excellent mechanical strength and hydrophobic properties. To evaluate its practicality, the composite underwent rigorous tests at a neighboring fish farm for a period of one year. The results indicated promising antifouling and effortless cleaning capabilities [136].

The viability of the aquaculture is contingent upon the antifouling efficacy of net cage. Future research should focus on accelerating the advancement of this technology. This objective can be accomplished by enhancing comprehension of the settlement and propagation of biofouling in the environment, investigating novel materials and antifouling coatings for infrastructure, and engineering breakthroughs to surmount the obstacles associated with antifouling technology [137].

4. Summary and outlook

The economic losses resulting from marine fouling organisms are significant, urging the need for effective antifouling technologies to guarantee the sustained utilization of marine equipment and the extraction of marine resources. The attachment of fouling organisms is influenced by various surface characteristics of the material, including surface free energy, morphology, wettability, and surface charge. (1) Bioactive antifoulant mechanisms, including alkaloids and organic acids, exhibit the capacity to induce microbial inactivation. (2) Self-polishing antifouling mechanisms possess the capability of generating dynamic renewable surfaces, thereby reducing the adhesion of fouling organisms. (3) Surface wettability mechanisms prevent the adhesion of fouling organisms through alterations in surface free energy or generation of hydration layer. (4) Photocatalytic antifouling mechanisms effectively decompose or transform fouling organisms by generating reactive oxygen species. (5) Biomimetic antifouling mechanisms demonstrate remarkable antifouling performance through the utilization of materials or structures inspired by natural phenomena. Simultaneously, the increasing demand for seawater resources has prompted a heightened focus on the implementation of biofouling control. It is

imperative to develop a cost-effective and efficient technology that combines excellent antifouling properties to sustainably extract resources such as uranium and lithium from seawater and enhance aquaculture efficiency. The marine antifouling strategies and antifouling systems for seawater resource utilization in detail are shown in Fig. 9 and Table 2.

Considering the complexity of marine environments, the challenge of developing cost-effective and versatile solutions for antifouling strategies persists in the field. Presently, antifouling mechanisms struggle to achieve a balance between stability, low toxicity, and affordability. It is important to acknowledge that a single antifouling strategy may not be universally applicable across all scenarios. Therefore, it is imperative to prioritize the development of antifouling technologies that incorporate synergetic strategies, drawing inspiration from natural phenomena. Fortunately, with the increasing comprehension of marine biofouling, there has been a heightened focus on the issue of fouling organisms in the utilization of seawater resources. To enhance the extraction efficacy of seawater resource, it is imperative to integrate efficient and environmentally benign antifouling strategies in future research and practical applications. We hope that the completeness of this review and its

explicit emphasis on seawater resource utilization, will inspire researchers to develop more efficient materials for the exploitation of seawater resources that are resistant to marine fouling organisms.

CRediT authorship contribution statement

Sheng Li: Writing – original draft. **Kang Feng:** Conceptualization, Writing – review & editing, Project administration. **Jiaying Li:** Writing – original draft. **Yan Li:** Conceptualization, Supervision, Writing – review & editing. **Zhengdong Li:** Writing – review & editing. **Liangmin Yu:** Conceptualization, Supervision, Writing – review & editing. **Xingtao Xu:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

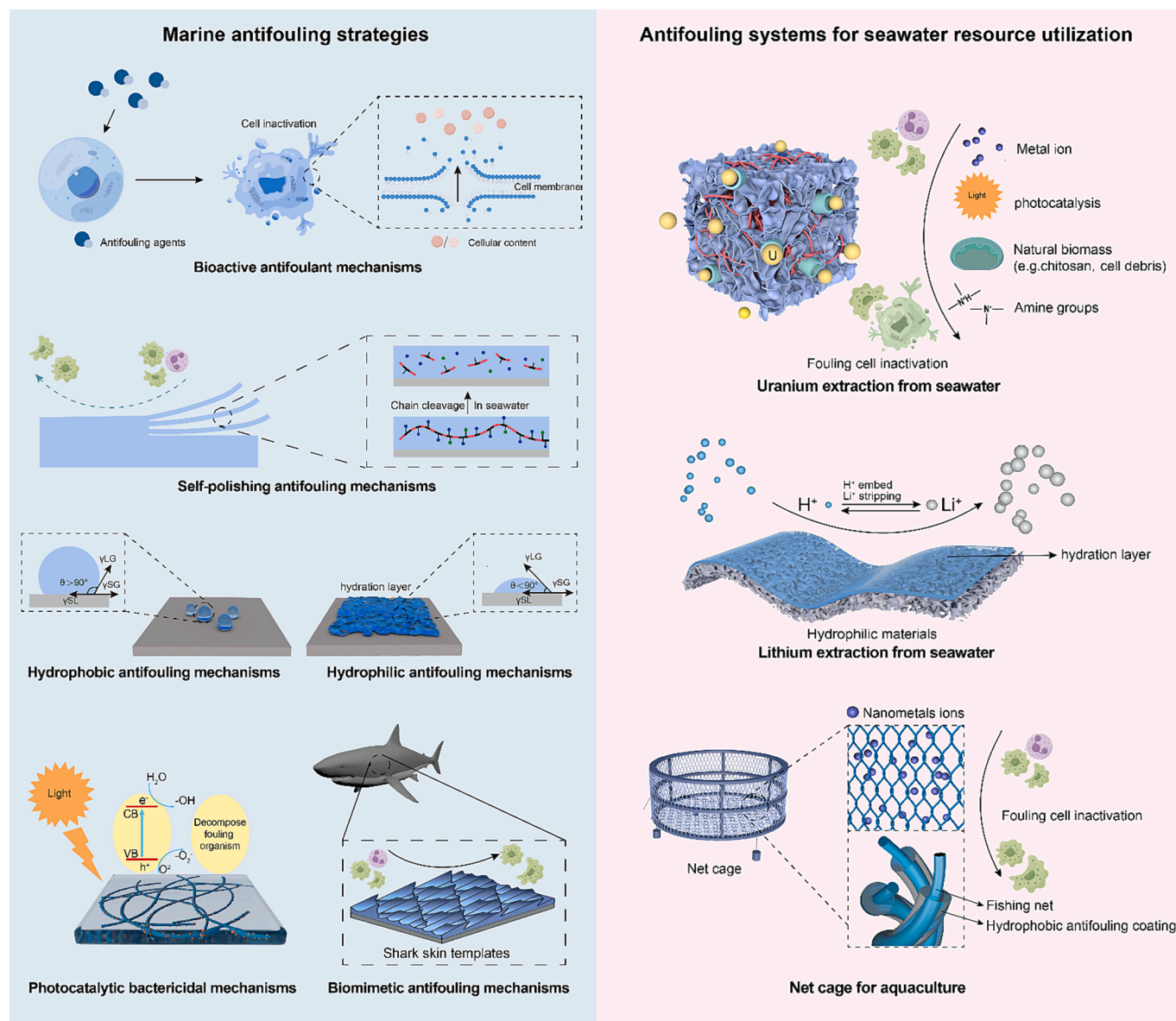


Fig. 9. Schematic diagram of marine antifouling strategies and antifouling systems for seawater resource utilization.

Table2

Marine antifouling strategies and antifouling systems for seawater resource utilization.

Antifouling strategies	Mechanisms of action	Ref.
Bioactive antifoulant mechanisms	Bioactive antifoulants possess the capability to readily permeate the cytoplasmic membrane of an organism, thereby inducing modifications to cellular integrity. At higher concentrations, these agents can lead to cellular lysis. Metallic antifoulants, such as Cu^{2+} , can cause fouling cell damage through electrostatic interactions between the positively charged metal and negatively charged cell membranes. Furthermore, metal ions have the potential to impede bacterial metabolic functions by binding to bacterial DNA. Certain natural antifoulants, such as trehalose lipid, induce cleavage through K^+ leakage, thereby altering the permeability of the cytomembrane to damage intact cells.	[138–140]
Self-polishing antifouling mechanisms	Upon application, the outer layer of the self-polishing surface interacts with seawater, initiating a hydrolysis and degradation reaction. These reactions lead to the cleavage of main chain and side chain of polymer, consequently transforming to small molecules or low molecular weight substances. The surface is polished through shedding caused by the shear force from seawater. The polishing rate is contingent upon the proportion of hydrophilic or seawater hydrolyzable groups and the composition of the copolymer.	[37]
Surface wettability mechanisms	Materials that exhibit a weak bond with fouling organisms have the capability to dislodge these organisms from their surface. This can be achieved through the shear force of water flow or mechanical removal, without the utilization of any toxic antifouling agent. The efficacy of hydrophobic antifouling mechanisms is contingent upon the material's low surface energy and low elastic modulus. Conversely, a surface with a hydrophilic polymer-induced hydration layer exhibits resistance against biofouling. The presence of a tightly bound hydration layer serves as a physical barrier and energy barrier, effectively inhibiting various processes including protein attachment, initial bacterial attachment, and subsequent biofilm formation. Amphiphilic coatings with hydrophobic and hydrophilic components not only provide a heterogeneous nanoscale mosaic chemical surface in which the	[52,65]

Table2 (continued)

Antifouling strategies	Mechanisms of action	Ref.
Photocatalytic bactericidal mechanisms	coexistence of hydrophobic and hydrophilic structural domains can also confuse organisms during sedimentation and adhesion processes. When exposed to ultraviolet (UV) light, photocatalytic antifouling mechanisms undergo a process where carriers such as electrons and holes are excited. Electrons are transferred from the valence band to the conduction band, while holes remain within the valence band. The electrons in the conduction band can be reduced to superoxide anion radicals, whereas the holes, possessing potent oxidizing ability, can facilitate the oxidation of water, leading to the production of hydroxyl radicals. The cell membranes of fouling organisms are susceptible to damage by radicals, resulting in cytoplasmic loss and subsequent cellular oxidation, ultimately leading to the death of an organism.	[73,74]
Biomimetic antifouling mechanisms	The surface microstructure of certain marine organisms (e.g., sharks) exhibits anti-biofouling properties by reducing adhesion between fouling organisms and surfaces. However, surface microstructure alone is inadequate in guaranteeing enduring antifouling effects. On the other hand, natural antifoulants from organisms are also an effective antifouling strategy. The corresponding mechanisms are similar to bioactive antifoulant mechanisms. Nevertheless, the specific and comprehensive mechanism underlying natural organism's antifouling capabilities remains uncertain.	[88]
Uranium extraction from seawater	Certain nano-metals, including Ti, Zn, Ag, exhibit potent antimicrobial properties by generating biotoxic ions. The porous adsorbent materials (COFs and MOFs) and metallic oxide possess photocatalytic capabilities, enabling the generation of hydroxyl radicals and reactive oxygen species (ROS) when exposed to light, thereby achieving antifouling effects. Certain organic groups, such as amine groups with a positive charge, present in polymer-based adsorbent materials can easily adhere to the surface of negatively charged microorganisms. These amine groups possess the capacity to interact with phosphate esters in the cell membranes or cell walls of fouling organisms via quaternary or tertiary ammonium salts, thereby compromising their cellular structure and energy	[106,107,111,121,122]

(continued on next page)

Table2 (continued)

Antifouling strategies	Mechanisms of action	Ref.
	metabolism. The abundant amine groups on natural biomass adsorbent materials such as chitosan and cell debris exhibit notable chemical activity, which can be modified by grafting, enabling them to chelate with metal ions and form hydrogen bonds with certain fouling organisms. Consequently, this facilitates the desired outcome of biofouling control.	
Lithium extraction from seawater	The hydrophilicity of the membrane enhances its fouling resistance by the formation of a tightly bound hydration layer on its surface, thereby improving the separation of lithium ions.	[128]
Net cage for aquaculture	The electrostatic interactions between positively charged nanometals and negatively charged cell membranes result in fouling cell damage. The hydrophobic coating on the outside of the net cages achieves biofouling control through low surface energy and low elastic modulus.	[135,136]

Data availability

No data was used for the research described in the article.

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